Carries, Rails and Resonance: the Structure of the Digit-Sum Parity Map

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Abstract

Let $s(n)$ be the sum of the decimal digits of n and let $f(n) = n + s(n)$ for odd n and $f(n) = n - s(n)$ for even n. Our main result concerns the 2-cycles of f, which are most of its observed cycles. We prove that an odd a lies on a 2-cycle if and only if $s(a)$ is odd and the addition $a + s(a)$ produces exactly $s(s(a))/9$ carries, and we show that this required number is within $1/9$ of the *mean* number of carries produced by adding $s(a)$ to a random odd integer (a "carry balance" identity). Combining this with exchangeability of digits, an anti-concentration lemma for carries and a local limit argument, we prove that the number of 2-cycles below N is of exact order $N/\sqrt{\log\log N}$. In particular 2-cycles have density zero, so their number does not grow linearly in N, although numerically they make up most of the observed cycles. We also obtain an asymptotic formula for the 2-cycle density with error $O((\log D)^2/\sqrt{D})$ below 10^D. Combining it with an exact formula for the variance of the carry count and a sharp local limit theorem, we prove an oscillation law: $\sqrt{\log_{10} D}\, T(D)/10^D$ has no limit, its limit points form the interval $[K_0\sqrt{726/301},\, K_0\sqrt{6}]$ with $K_0 = 1/(12\sqrt{11\pi})$, and the peak-to-trough ratio is $\sqrt{301}/11 \approx 1.577$.

Along the way we establish several exact structural facts: on multiples of nine, f is conjugate to $m \mapsto m \pm \kappa(m)$ with $\kappa(m) = s(9m)/9$ given by a digit formula; a general level equation describes how the digit sum changes under each step; a *conditional* time-reversal identity $f(f(x) + 9) = x + 9$ holds whenever $s(x)$ is even and $s(f(x) + 9) = s(x)$; and the orbit of $H \cdot 10^L + x$ depends on H only through $s(H)$ while it stays in one prefix block, which yields a cycle-counting algorithm with a rigorous error bound. We prove that an orbit is eventually periodic if and only if its digit sums are bounded, and that every run of the orbit along which the digit sum stays constant moves it by at most $O(k^2)$, where $9k$ is that digit sum. Using this and prefix lifting we give a computer-assisted proof that every orbit starting below 10^{341} is eventually periodic (and, for the binary analogue, every orbit starting below 2^{3024}). Whether *every* orbit is eventually periodic remains open; we reduce it to a conjectured quadratic bound on climbs of finite auxiliary maps.

Finally we treat every base b. In odd bases every orbit reaches 0. In every even base, including base 2 where the carry mechanism takes a different form, the structural results carry over and the 2-cycles again have density of exact order $(\log D)^{-1/2}$, with resonances at the base-b repunits $b^j - 1$.

1 Introduction

Iterations built from the digits of an integer have a long history. The sequence $n \mapsto n + s(n)$ ("digit addition", or "digital rivers") and the associated self numbers go back to Kaprekar [13] and are catalogued in the OEIS [20, 16]; the trajectories of this map always move upwards. Several parity-sensitive digit maps that can return have been recorded in the OEIS, mostly as lists of their periodic points; Table 1 compares them with ours.

map	rule	what is recorded
digital river [20]	$n \mapsto n + s(n)$	orbits, self numbers [16]
A124176 [17]	$m \mapsto m +$ (sum of odd digits) − (sum of even digits)	periodic points
A124177 [18]	$m \mapsto m +$ (sum of even digits) − (sum of odd digits)	periodic points
A243260 [19]	$n \mapsto n \pm s(n)$, sign by the parity of $s(n)$	numbers that recur
this paper	$n \mapsto n \pm s(n)$, sign by the parity of n	structure, densities, convergence

Table 1: Neighbouring digit maps. In A124176/A124177 "odd digits" are digits with odd value, not digits in odd positions.

The maps of A124176 and A124177 move by a signed digit sum that depends on the digit values, not on a global parity, so they have no analogue of our rails. In A243260 the branch is chosen by the parity of the digit sum. That agrees with our rule in odd bases, where $n \equiv s_b(n) \pmod 2$ (Section 11), but not in base 10. Branching on the parity of the *number itself* produces the two rails of Section 4, and with them the carry characterisation of 2-cycles. We have not found structural or asymptotic results for any of these neighbouring maps; the OEIS entries list terms and give no proofs about densities or convergence. This is a targeted rather than an exhaustive search, so the novelty claim is for the results below, not for the idea of parity-dependent digit maps. The number of carries in random addition is a well-studied stochastic process: Holte found the remarkable transition matrix of the carries Markov chain [12], Diaconis and Fulman connected it with riffle shuffles and symmetric functions [11], Borodin, Diaconis and Fulman placed it among one-dependent determinantal processes [9], and Diaconis, Shao and Soundararajan studied carries from the viewpoint of group theory and additive combinatorics [10].

Maps that branch on the parity of n are a natural test bed for deterministic dynamics that behave "as if random": the branching couples a local, arithmetic rule (here the propagation of carries) to a global dichotomy, and the resulting orbits are often hard to control. The $3x + 1$ map [15] is the best-known example; we mention it only as background, and do not suggest that the map studied here is comparably difficult. Indeed, the point of this paper is that for the map below the branching interacts with carries in a way that can be analysed exactly. We study

$$f(n) = \begin{cases} n + s(n), & n \text{ odd}, \\ n - s(n), & n \text{ even}, \end{cases} \qquad n \geq 0, \tag{1}$$

the *digit-sum parity map*; the parity condition refers to the input n, not to its digit sum. We have not found earlier work on this particular map; its novelty lies less in the definition than in the connection it creates between a deterministic integer iteration and the probability theory of carries.

Why this map? It is the simplest parity-branching map built from the digit sum: the two branches are the digit-addition map and its inverse direction. This choice makes it a minimal model in which a deterministic dynamical question can be translated *exactly* into a question about carries. Its 2-cycles are characterised by a carry count (Theorem 6.2), that count is pinned near its mean by a balance identity (Theorem 10.1), and the density of periodic points then becomes a local-limit problem for the carries chain (Theorem 10.5). We expect the same mechanism to govern other parity-branching digit maps, for instance in other bases or with weighted digit sums, and view the present map as a test case for it.

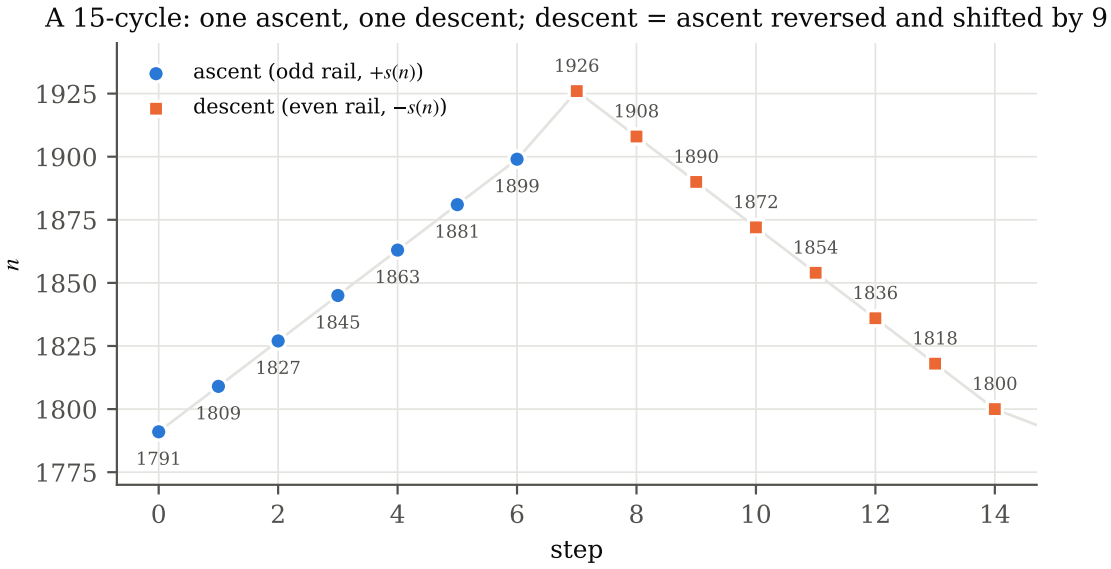


Figure 1: The 15-cycle through 1791. Odd terms (circles) ascend by 18 (once by 27); even terms (squares) descend by 18 (once by 9). Here the descent is the ascent run backwards and shifted by +9, as predicted by the conditional reversibility identity (Theorem 5.1).

Numerically every orbit enters a short cycle after a few dozen steps at most, usually a 2-cycle such as $27 \leftrightarrow 36$. Figure 1 shows a typical longer cycle: a regular ascent followed by a regular descent that retraces it.

Main result. Let $T(D)$ be the number of odd integers below 10^D that lie on a 2-cycle.

Main Theorem (Theorems 6.2, 9.4 and 10.5)**.** *An odd a lies on a 2-cycle if and only if $s(a)$ is odd and $a + s(a)$ produces exactly $s(s(a))/9$ carries. There are absolute constants $c, K > 0$ such that, for all sufficiently large D,*

$$\frac{c}{\sqrt{\log D}} \le \frac{T(D)}{10^D} \le \frac{K}{\sqrt{\log D}}.$$

The mechanism is simple to state. The number of carries required for a 2-cycle is essentially its own mean (Theorem 10.1); the carry count is a sum of about $\log D$ weakly dependent terms; and such a sum hits a value at its centre with probability of order $(\log D)^{-1/2}$.

Further results.

- Self-similarity on multiples of nine and a digit formula for $s(9m)$ (Theorem 3.1).
- A general level equation describing how the digit sum changes under each step, and the resulting "rail" picture (Theorem 4.2).
- A conditional time-reversal identity (Theorem 5.1); in the computed range, 43,194 of the 43,210 nonzero cycles have the "mountain" shape that it suggests. We do not prove a corresponding statement for all cycles.

- Prefix lifting (Theorem 7.1): an exact count of cycles confined to one prefix block, and a count of all cycles below 10^D with an explicit error of at most $18D \cdot 10^{D-L}$ (Corollary 7.2).
- Carry resonance: an asymptotic formula for $T(D)/10^D$ (Theorem 9.7); an exact formula for the variance of the carry count, $\frac{11}{9}\sum_i q_i(1-q_i)$ (Lemma 10.6); a sharp local limit theorem for carries (Theorem 10.9); and the resulting *oscillation law* (Theorem 10.10): the normalised density $\sqrt{\log_{10} D}\, T(D)/10^D$ has no limit, its limit points fill an explicit interval, and $\limsup / \liminf = \sqrt{301}/11$ (Corollary 10.11).
- A convergence criterion: an orbit is eventually periodic if and only if its digit sums are bounded (Theorem 12.2); and every constant-level run moves an orbit by $O(k^2)$ (Theorem 12.5); and a computer-assisted proof that every orbit starting below 10^{341} (below 2^{3024} in base 2) is eventually periodic (Theorems 12.6 and 12.7). Global eventual periodicity (Conjecture 12.10) remains open.
- Other bases (Section 11): in odd bases every orbit reaches 0 (Theorem 11.1); in every even base the structure theorems hold (Theorem 11.2) and the 2-cycle density is of exact order $(\log D)^{-1/2}$ (Theorem 11.5); in base 2 this requires a different switch mechanism (Lemma 11.4).

2 Preliminaries

Table 2 collects the notation used throughout.

symbol	meaning
$s(n)$, $s_b(n)$	digit sum of n in base 10, base b
f, f_b	the digit-sum parity map (1) and its base-b analogue
$\mathrm{C}(a,b)$, $\mathrm{B}(a,b)$	number of carries in $a+b$, of borrows in $a-b$
$\mathrm{C}_K(a,t)$	carries out of columns $0,\dots,K-1$ only
k, level	$s(n)=9k$ for a multiple of 9; $\kappa(m)=s(9m)/9$
$T(D)$	number of odd $a<10^D$ lying on a 2-cycle
$P_D(t)$	probability that D uniform digits sum to t
$\rho(t)$	probability that adding t to a random odd number gives exactly $s(t)/9$ carries
$S(t)$, W	switch positions of t (Lemma 9.1) and the random set of active switches
q_i, φ	$q_i=\Pr(\text{carry into column } i)$; $\varphi(x)=x(1-x)$ (Section 10.1)
h, B_D, K_0	scales and constants of the oscillation law (Theorem 10.10)
H, L, σ	prefix and block length in $n=H10^L+x$; $\sigma=s(H)$
g_σ, $\bar{g}_\sigma$	low map on $[0,b^L)$ and on the half-line (Section 12)
$c_L(\sigma)$	number of cycles of g_σ confined to one block (Section 7)

Table 2: Notation.

All numbers are non-negative integers written in base 10, with digits $n=\sum_i n_i 10^i$. For $a,b\ge 0$ let $\mathrm{C}(a,b)$ be the number of carries in the schoolbook addition $a+b$ (counting a final carry into a new leading digit), and for $a\ge b\ge 0$ let $\mathrm{B}(a,b)$ be the number of borrows in $a-b$.

Lemma 2.1 (Carries and borrows)**.** *$s(a+b)=s(a)+s(b)-9\,\mathrm{C}(a,b)$, and for $a\ge b$, $s(a-b)=s(a)-s(b)+9\,\mathrm{B}(a,b)$.*

Proof. With incoming carries $c_i \in \{0,1\}$, $c_0 = 0$, column i produces the digit $a_i + b_i + c_i - 10c_{i+1}$, and a final carry produces a leading digit equal to it. Summing, $s(a+b) = s(a)+s(b)-9\sum_i c_{i+1} = s(a) + s(b) - 9\,\mathrm{C}(a,b)$. The second identity is the first applied to $(a-b)+b = a$, whose carries are the borrows of $a - b$. □

This is the base-10 form of the identity behind Kummer's theorem on carries [14]; the statistics of carries are studied in [12, 11].

Lemma 2.2. *$s(n) \equiv n \pmod 9$; hence $9 \mid f(n)$ for even n, and $9 \mid n$ implies $9 \mid f(n)$.*

Proof. Since $10 \equiv 1 \pmod 9$, $n = \sum_i n_i 10^i \equiv \sum_i n_i = s(n) \pmod 9$. For even n, $f(n) = n - s(n) \equiv 0 \pmod 9$. If $9 \mid n$ then $9 \mid s(n)$, so $9 \mid n \pm s(n) = f(n)$. □

Proposition 2.3 (Reduction to multiples of nine)**.** *Every term after the first even term of an orbit is a multiple of 9. The first even term may come late: $11 \to 13 \to 17 \to 25 \to 32 \to 27$ needs five steps to reach a multiple of 9, and among $n < 10^6$ the maximum is 14 steps (at $n = 899457$).*

Proof. If m is even then $9 \mid f(m)$ by Lemma 2.2, and by the same lemma every later term is again a multiple of 9, by induction. The statements about 11 and about $n < 10^6$ are computational: they were obtained by iterating f from every starting value (see the Data and code section). □

Remark 2.4. In particular, an orbit need not reach a multiple of nine within two steps, and an odd multiple of 9 need not map to an even number: if its digit sum is even it stays odd, and this is exactly what produces the ascending runs of Section 4.

3 Self-similarity and a digit formula for $s(9m)$

Theorem 3.1 (Self-similarity)**.** *For $m \geq 0$ let $\kappa(m) = s(9m)/9$. Then*

$$\kappa(m) \;=\; \mathrm{B}(10m, m) \;=\; \#\big\{\, i \geq 0 :\; m_i > 9 \cdot 0.m_{i-1}m_{i-2}\cdots m_0 \,\big\},$$

where $0.m_{i-1}\cdots m_0$ is the number in $[0,1)$ formed by the digits below position i (empty for $i = 0$). Moreover $9m \mapsto f(9m)$ is conjugate to

$$g(m) = \begin{cases} m + \kappa(m), & m \text{ odd}, \\ m - \kappa(m), & m \text{ even}. \end{cases}$$

Proof. Since $9m = 10m - m$ and $s(10m) = s(m)$, Lemma 2.1 gives $s(9m) = s(10m) - s(m) + 9\,\mathrm{B}(10m, m) = 9\,\mathrm{B}(10m, m)$. In a subtraction $a - b$ a borrow leaves column i exactly when $a \bmod 10^{i+1} < b \bmod 10^{i+1}$. With $a = 10m$, $a \bmod 10^{i+1} = 10\,(m \bmod 10^i)$, and the condition $10(m \bmod 10^i) < m_i 10^i + (m \bmod 10^i)$ is $m_i > 9(m \bmod 10^i)/10^i$. Finally $9m$ and m have the same parity, so $f(9m) = 9m \pm s(9m) = 9\,g(m)$. □

For instance $\kappa(11) = 2$: $m_0 = 1 > 0$ and $m_1 = 1 > 9 \cdot 0.1$; indeed $s(99) = 18$. The theorem was checked against direct computation for all $m < 10^6$.

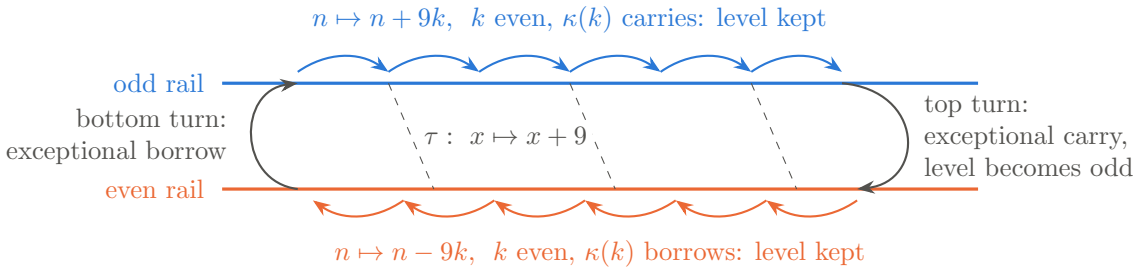


Figure 2: Rails and turns. Regular steps translate along a rail; the shift τ maps the odd rail to the even rail and reverses time (Theorem 5.1), so the descent retraces the ascent.

4 Rails and the level equation

Definition 4.1. On multiples of 9 write $s(n) = 9k$ and call k the *level* of n. The *odd rail* consists of the odd multiples of 9 and the *even rail* of the even ones.

Theorem 4.2 (Level equation)**.** *Let $9 \mid n$ with level k, i.e. $s(n) = 9k$, and let $\kappa(k) = s(9k)/9$ as in Theorem 3.1. Then*

$$s(n + 9k) = 9\big(k + \kappa(k) - \mathrm{C}(n, 9k)\big), \qquad s(n - 9k) = 9\big(k - \kappa(k) + \mathrm{B}(n, 9k)\big) \quad (n \geq 9k).$$

Thus an ascending step keeps the level if and only if it produces exactly $\kappa(k)$ carries, and a descending step keeps the level if and only if it produces exactly $\kappa(k)$ borrows. A step keeps the rail (the parity of n) if and only if k is even. In particular, for $1 \leq k \leq 10$ one has $\kappa(k) = 1$, so the level is kept exactly when there is one carry or one borrow.

Proof. By Lemma 2.1, $s(n \pm 9k) = s(n) \pm s(9k) \mp 9$ C (resp. ± 9 B), and $s(9k) = 9\kappa(k)$. The parity of $n \pm 9k$ equals that of n iff $9k$ is even, i.e. k is even. For $1 \leq k \leq 10$, $9k \in \{9, 18, \ldots, 90\}$ has digit sum 9. □

The general form matters because a typical D-digit multiple of 9 has level $k \approx D/2$, far beyond 10 when D is large; the equation was checked on 20,000 random multiples of 9 with up to 60 digits, 13,315 of them with $k > 10$. Theorem 4.2 explains the regular motion seen in Figure 1: for small k, adding $9k = 10k - k$ typically lowers the units digit by k and raises the tens digit by k, leaving the digit sum and hence the step size unchanged. In general the orbit is a translation along its rail as long as each step produces exactly $\kappa(k)$ carries or borrows; an *exceptional* step, with a different number, changes the level. Figure 2 summarises the resulting picture.

5 A conditional time-reversal identity and mountain cycles

Theorem 5.1 (Conditional reversibility)**.** *Let $x \geq 0$. If $s(x)$ is even and $s(f(x) + 9) = s(x)$, then*

$$f\big(f(x) + 9\big) = x + 9.$$

Equivalently, writing $\tau(x) = x+9$, we have $f(\tau(f(x))) = \tau(x)$ at every such x. This is a conditional identity, not a global symmetry: it says nothing about points where either hypothesis fails.

Proof. Let $S = s(x)$, even. If x is odd, $f(x) = x + S$ is odd and $y = x + S + 9$ is even, so $f(y) = y - s(y) = x + S + 9 - S = x + 9$. If x is even, $f(x) = x - S$ is even and $y = x - S + 9$ is odd, so $f(y) = y + s(y) = x + 9$. □

When the hypotheses hold along a whole stretch of an orbit, the stretch is retraced, shifted by 9, in the opposite direction. They hold on *regular* stretches: by Lemma 2.1, $s(f(x)+9) = s(f(x))+9-9\,\mathrm{C}(f(x),9)$, which equals $s(f(x))$ exactly when adding 9 to $f(x)$ produces one carry, while $s(f(x)) = s(x)$ on a level-preserving step (Theorem 4.2). The identity was also checked directly for all 31,196 applicable $x < 10^6$.

This suggests the *mountain* shape seen in Figure 1: an ascent on the odd rail, a turn, a descent on the even rail that is the τ-image of the ascent run backwards, and a second turn that closes the loop. We record its prevalence as a computational statement only.

Proposition 5.2 (Computational census)**.** *Of the* 43,210 *distinct cycles other than the fixed point* $\{0\}$ *reached from the starting values* $1 \le n \le 2 \cdot 10^6$ *(the fixed point* 0*, reached for instance by* $1 \to 2 \to 0$*, brings the total to* 43,211*), exactly* 43,194 *consist of one ascending run (odd terms) followed by one descending run (even terms); the other* 16 *have two ascending and two descending runs, the smallest being* $799020 \to 798993 \to 799038 \to 799002 \to 798975$*.*

We do not know whether almost all cycles of f, in the sense of density, have the mountain shape; Theorem 5.1 gives a mechanism for it but not a proof. A 2-cycle is a mountain without regular stretches.

6 Two-cycles

Proposition 6.1 (Two-step identity)**.** *If n and $s(n)$ are odd, then $f(n)$ is even and $f^2(n) = n - s(s(n)) + 9\ \mathrm{C}(n, s(n))$.*

Proof. By Lemma 2.1, $s(f(n)) = s(n) + s(s(n)) - 9\,\mathrm{C}(n, s(n))$, and $f^2(n) = n + s(n) - s(f(n))$. □

Theorem 6.2 (Characterisation of 2-cycles)**.** *Every 2-cycle $\{a, b\}$, $a \neq b$, contains exactly one odd number, and an odd a lies on a 2-cycle if and only if $s(a)$ is odd and $s(s(a)) = 9\ \mathrm{C}(a, s(a))$. Then $9 \mid a$; and if $a < 10^{11} - 1$ the condition reads: $s(a)$ is odd and $a + s(a)$ has exactly one carry.*

Proof. Two even (or two odd) points would give $b < a < b$. For odd a, $b = a + s(a)$ must be even (else $f(b) > b > a$), so $s(a)$ is odd, and $f(b) = a$ is $f^2(a) = a$, i.e. $s(s(a)) = 9\,\mathrm{C}(a, s(a))$ by Proposition 6.1. The converse is immediate. Since $s(s(a)) \equiv a \pmod 9$, $9 \mid a$. For $a < 10^{11} - 1$, $s(a) \le 98$, so $s(a) \in \{9, 27, 45, 63, 81\}$, each with digit sum 9. □

Example 6.3. $27 + 9 = 36$ has one carry, so $\{27, 36\}$ is a 2-cycle. $999 + 27 = 1026$ has three, so 999 is not on a 2-cycle: $f(1026) = 1017$.

Let $T(D)$ be the number of odd $a < 10^D$ on a 2-cycle. By Theorem 6.2, $T(D)$ is computed exactly by a digit-by-digit dynamic programme over (carry, digit sum so far, carries so far), summing over admissible $t = s(a) \equiv 9 \pmod{18}$; it agrees with exhaustive enumeration for $D \le 5$. Values are given in Table 3.

7 Prefix lifting

Theorem 7.1 (Prefix lifting)**.** *Fix $L \ge 1$ and for $\sigma \ge 0$ let $g_\sigma(x) = x + \sigma + s(x)$ for odd x and $x - \sigma - s(x)$ for even x. Let $c_L(\sigma)$ be the number of cycles of g_σ contained in $[0, 10^L)$. Then for every prefix H the cycles of f all of whose terms have the form $H \cdot 10^L + x$ with $0 \le x < 10^L$*

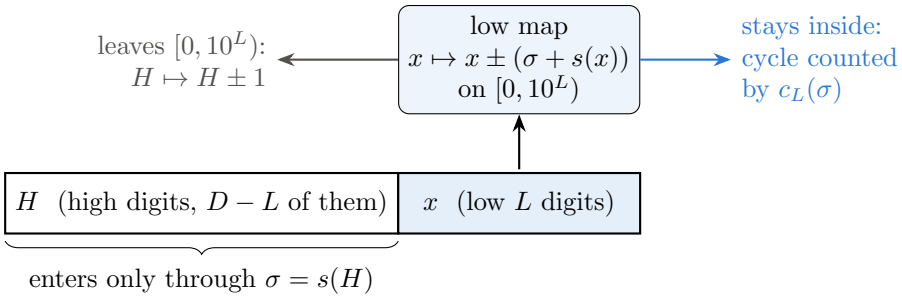


Figure 3: Prefix lifting. Since $s(n) = s(H) + s(x)$ and n has the parity of x, the orbit of $H \cdot 10^L + x$ depends on H only through $\sigma = s(H)$ until the prefix changes.

correspond bijectively to these cycles of $g_{s(H)}$. Consequently the number of cycles of f below 10^D whose terms share a common prefix (above the last L digits) is

$$\sum_{\sigma \geq 0} c_L(\sigma)\, N(D - L, \sigma), \qquad N(d, \sigma) = [z^\sigma]\Big(\frac{1 - z^{10}}{1 - z}\Big)^d,$$

where $N(d, \sigma)$ counts d-digit strings (leading zeros allowed) with digit sum σ. The same holds for cycles of any fixed length.

Proof. For $n = H \cdot 10^L + x$ with $0 \leq x < 10^L$ we have $s(n) = s(H) + s(x)$, and $n \equiv x \pmod 2$. So as long as $g_{s(H)}(x) \in [0, 10^L)$, $f(n) = H \cdot 10^L + g_{s(H)}(x)$. A cycle of f with constant prefix H is therefore a cycle of $g_{s(H)}$ inside $[0, 10^L)$, and conversely. Summing over prefixes $H < 10^{D-L}$ grouped by $s(H)$ gives the formula. □

Theorem 7.1 counts only cycles confined to one prefix block. The remaining cycles cross a block boundary, and they are few.

Corollary 7.2 (Counting all cycles, with error bound)**.** *Let $Z(D)$ be the number of cycles of f all of whose terms are below 10^D (including the fixed point 0), and let $Z_L(D) = \sum_\sigma c_L(\sigma) N(D - L, \sigma)$. If $9D < 10^L$, then*

$$Z_L(D) \;\leq\; Z(D) \;\leq\; Z_L(D) + 18D \cdot 10^{D-L}.$$

The same bound holds for cycles of any fixed length, in particular for 2-cycles.

Proof. Every cycle counted by $Z_L(D)$ is counted by $Z(D)$. A cycle not counted by $Z_L(D)$ has terms in at least two prefix blocks, so it contains a term n with $f(n)$ in a different block. Since $|f(n) - n| = s(n) \leq 9D < 10^L$, the low part $x = n \bmod 10^L$ then satisfies $x \geq 10^L - 9D$ (if $f(n) > n$) or $x < 9D$ (if $f(n) < n$). At most $2 \cdot 9D \cdot 10^{D-L}$ integers below 10^D have such a low part, and distinct cycles are disjoint. □

Algorithm. Compute $c_L(\sigma)$ for all needed σ (a functional graph on 10^L vertices per σ, with cycles found by pointer doubling), then evaluate the polynomial coefficients. This costs $O(10^L \cdot L \cdot \sigma_{\max} + D\,\sigma_{\max})$ with $\sigma_{\max} \leq 9D$, instead of the 10^D orbits of brute force, and by Corollary 7.2 it determines $Z(D)/10^D$ to within $18D/10^L$. Choosing L a little larger than $\log_{10} D$ therefore makes the relative error as small as desired. For example, for $D = 200$ the guaranteed error in $Z(D)/10^D$ is 0.36 with $L = 4$, 0.036 with $L = 5$ and 0.0036 with $L = 6$. The profile costs about $10^L \cdot L$ operations for each of the $\sigma_{\max} \approx 1{,}800$ values of σ, i.e. about 10^9 operations for $L = 5$; in our

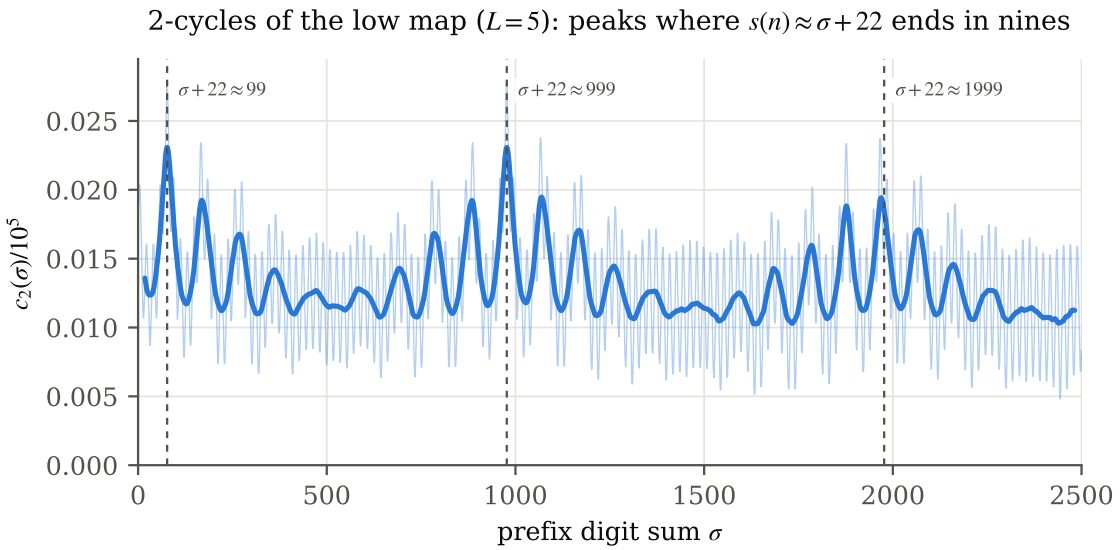


Figure 4: Number of 2-cycles of the low map g_σ for $L = 5$ (thin) and an 18-term moving average (thick). Peaks occur where the typical digit sum $s(n) \approx \sigma + 22$ ends in a run of nines.

implementation the $L = 5$ profile (for all $\sigma \le 2{,}500$) took under a minute on a laptop, and each extra digit of L multiplies the cost by about 10. In practice the agreement is much better than this bound: with $L = 3$ the formula reproduces the brute-force counts 2026 ($D = 5$) and 21566 ($D = 6$) exactly, and with $L = 5$, restricted to 2-cycles, it reproduces the exact $T(D)/10^D$ to six decimal places for every $6 \le D \le 200$ in Table 3 (where the bound $18D/10^L$ alone would only guarantee about two decimal places).

The profile $c_2(\sigma)$ (Figure 4) is irregular, and that is the source of every oscillation in the counts: the density of cycles below 10^D is the average of $c_L(\sigma)/10^L$ against the digit-sum distribution of a random $(D - L)$-digit prefix, a bell curve centred at $4.5(D - L)$ of width about $\sqrt{8.25D}$, which slides across the irregularities as D grows.

8 Carry resonance and the density of 2-cycles

For an odd multiple t of 9 let $\rho(t)$ be the probability that adding t to a random integer with odd units digit produces exactly $s(t)/9$ carries. With a uniformly random digit of a, a column whose digit of t is d and whose incoming carry is c produces a carry with probability $(d + c)/10$. Hence, marking carries by z,

$$\rho(t) = [z^{s(t)/9}]\; u^{\top} M_1(t_0)\, M(t_1) \cdots M(t_{\ell-1})\, M(0)^{\infty}\, \mathbf{1}, \qquad M(d) = \frac{1}{10}\begin{pmatrix} 10-d & d\,z \\ 9-d & (d+1)\,z \end{pmatrix},$$

with $u = (1, 0)^{\top}$ and M_1 the units column restricted to odd digits. This is the carries Markov chain [12, 11].

Proposition 8.1. *$\rho(10^j - 1) = 9/10$ for every $j \ge 1$.*

Proof. Adding $t = 10^j - 1$ to a with odd units digit: the units column gives $a_0 + 9 \ge 10$, and each of the next $j - 1$ columns gives $a_i + 9 + 1 \ge 10$, so all j columns carry and $s(t)/9 = j$. Exactly j carries occur iff column j does not carry, i.e. $a_j \ne 9$, which has probability $9/10$. □

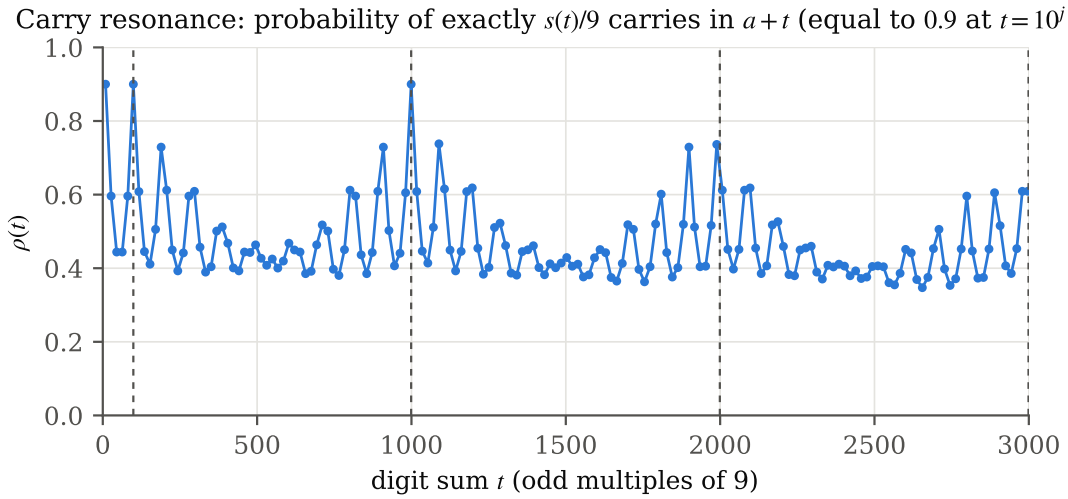


Figure 5: $\rho(t)$ for odd multiples t of 9. The maximum 9/10 is attained at $t = 10^j - 1$ (Proposition 8.1).

D	exact	resonance	D	exact	resonance
6	0.018879	0.017810	30	0.013760	0.013711
8	0.013812	0.014153	50	0.013827	0.013807
10	0.013101	0.012902	75	0.012426	0.012412
12	0.012434	0.012450	100	0.011919	0.011919
15	0.013855	0.013795	150	0.012165	0.012163
20	0.019908	0.019996	200	0.014819	0.014821

Table 3: Proportion $T(D)/10^D$ of integers below 10^D that are the odd point of a 2-cycle: exact (digit dynamic programme) and resonance approximation.

When t consists of nines the required number of carries is forced; when t has small or mixed digits, matching it is a coin flip (for example $\rho(45) = \rho(63) = 0.444$). Figure 5 shows the resulting *carry resonance*.

Treating $s(a) = t$, the parity of a and the carries as independent gives the *resonance approximation*

$$\frac{T(D)}{10^D} \;\approx\; \frac{1}{2} \sum_{t \equiv 9 \pmod{18}} P_D(t)\, \rho(t),$$

where $P_D(t)$ is the probability that a random D-digit string has digit sum t. Table 3 compares it with the exact values: the error is below 10^{-4} for $12 \le D \le 200$ and below $2 \cdot 10^{-6}$ for $100 \le D \le 200$.

Since $s(a)$ concentrates near $4.5D$, the density peaks when $4.5D$ lies just below a power of ten (the relevant t end in nines) and dips in between. Evaluating the approximation up to $D = 10^4$ (Figure 6) gives peaks near $D = 22, 218, 2247$ with heights 0.0210, 0.0160, 0.0145, and troughs near $D = 554$ and 5452 with values 0.0109 and 0.0098; at $D = 10^4$ the density is about 0.0096.

Conjecture 8.2 (now Corollary 10.11)**.** *The normalised density* $\sqrt{\log D}\, T(D)/10^D$ *has no limit as* $D \to \infty$*, with local maxima near* $D \approx (10^j - 1)/4.5$.

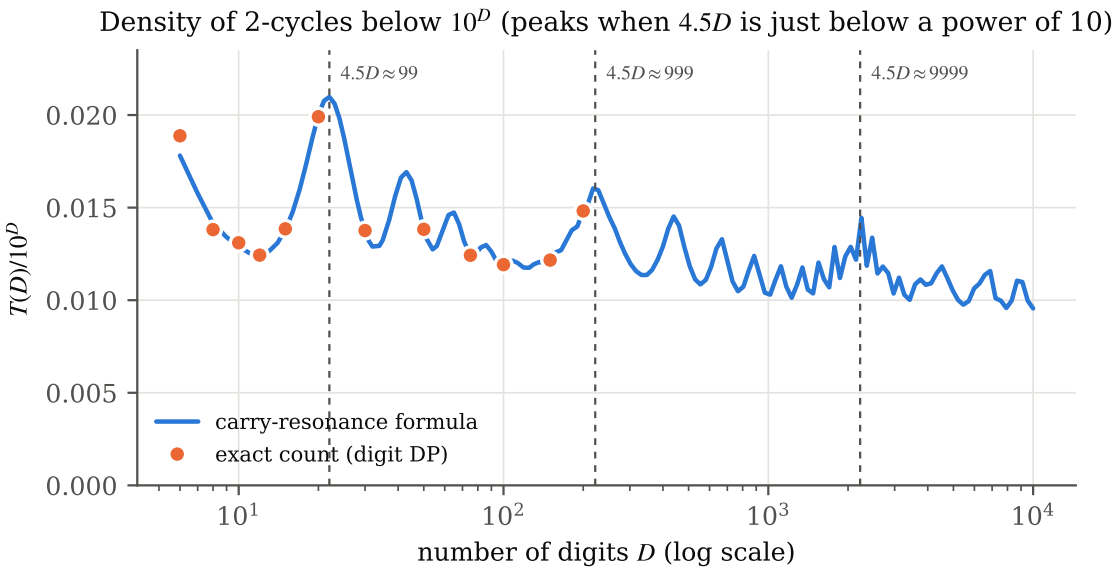


Figure 6: Density of 2-cycles below 10^D: exact values (dots) and the resonance approximation (line), with $4.5D = 99, 999, 9999$ marked.

We first stated this from the data in Table 4, which suggested a peak-to-trough ratio of about 1.4. Corollary 10.11 proves it and identifies the limit points exactly. The ratio $\limsup / \liminf$ equals $\sqrt{301}/11 \approx 1.577$; the smaller values in the table are a pre-asymptotic effect.

By Theorem 9.7, the conjecture concerns only the main term $\frac{1}{2}\sum_t P_D(t)\rho(t)$, since its error $O((\log D)^2/\sqrt{D})$ is much smaller than the main term, which is of order $(\log D)^{-1/2}$. We evaluated the main term with a Gaussian approximation to P_D (which agrees with the exact convolution to six decimal places at $D = 200, 1000, 5000$) up to $D \approx 6 \cdot 10^7$. Table 4 shows that the ratio between consecutive local maxima and minima of the normalised density stays between 1.375 and 1.413 over six orders of magnitude of D.

j	peak at D	normalised peak	trough at D	normalised trough	ratio
3	222	0.03766	777	0.02702	1.394
4	2,222	0.04128	7,777	0.02947	1.401
5	22,222	0.04178	38,888	0.03037	1.375
6	222,222	0.04387	614,035	0.03144	1.395
7	2,222,222	0.04406	6,345,029	0.03160	1.394
8	22,222,222	0.04545	63,450,292	0.03217	1.413

Table 4: Local maxima and minima of $\sqrt{\ln D} \cdot \frac{1}{2}\sum_t P_D(t)\rho(t)$ (computational; the extrema were located by a search over a grid of D near $(10^j - 1)/4.5$ and $2.5 \cdot 10^j/4.5$).

The mechanism is the one behind Proposition 8.1. The digit sum t of a random D-digit number lies within $O(\sqrt{D})$ of $4.5D$, so its top $\approx \frac{1}{2}\log_{10} D$ digits are determined by D and only its bottom digits vary. When $4.5D$ is just below 10^j, those top digits are nines; columns of $a + t$ where t has a nine carry almost deterministically, so only the bottom half of the columns contributes to the variance of the carry count, and the probability of hitting its mean is correspondingly larger. At

other values of D all columns contribute. Section 10.1 makes this precise. The variance of the carry count equals $\frac{11}{9}\sum_i \varphi(q_i)$ exactly (Lemma 10.6), and a sharp local limit theorem (Theorem 10.9) turns it into the oscillation law, Theorem 10.10.

9 Two-cycles have density zero

Throughout this section $K, K_0, K_1, \dots$ denote absolute constants and $c > 0$ an absolute constant, not necessarily the same at each occurrence. For an integer $t \geq 1$ with digits t_i let

$$S(t) = \{\, i \text{ odd},\ i \geq 1 :\ t_i \in \{1, \dots, 8\} \,\}.$$

For a random variable X and $\lambda > 0$, $Q(X;\lambda) = \sup_x \Pr(x \leq X \leq x+\lambda)$ is its concentration function. We use the Kolmogorov–Rogozin inequality [8, 7]: for independent $X_1, \dots, X_n$ and $0 < \lambda \leq L$,

$$Q\Big(\sum_i X_i; L\Big) \leq K_0\, L\Big(\sum_i \lambda^2\big(1 - Q(X_i;\lambda)\big)\Big)^{-1/2}. \tag{2}$$

For integer-valued variables and $\lambda = L = \frac{1}{2}$, $Q(X;\frac{1}{2})$ is the largest point probability, so (2) gives $\max_m \Pr(\sum_i X_i = m) \leq K_0\big(\sum_i (1 - \max_x \Pr(X_i = x))\big)^{-1/2}$.

Lemma 9.1 (Switch lemma)**.** *Let $t \geq 1$ have ℓ digits, let $0 < \delta \leq \frac{1}{10}$, and let $a = \sum_i a_i 10^i$ have independent digits $a_0, a_1, \dots$ such that $\Pr(a_i = d) \geq \delta$ for all $d \in \{0, \dots, 9\}$ and $1 \leq i \leq \ell$. Let C be the number of carries in $a + t$, possibly counting only the carries out of columns $0, \dots, k-1$ for some fixed $k > \ell$. Then for every integer m*

$$\Pr(C = m) \ \leq\ \frac{K_1}{\delta\sqrt{|S(t)|}} + e^{-9\delta|S(t)|/8}.$$

Proof. Write c_j for the carry into column j. For $i \in S(t)$ call i a *switch* if $a_{i+1} + t_{i+1} \neq 9$, and let W be the set of switches. Since $i+1$ is even, W depends only on the digits at even positions, and the events $\{i \in W\}$, $i \in S(t)$, are independent with $\Pr(i \in W) = 1 - \Pr(a_{i+1} = 9 - t_{i+1}) \geq 9\delta$.

Condition on $G = (a_j)_{j \notin W}$, which determines W; the digits a_i, $i \in W$, remain independent with their original laws. We claim that

$$C = C_0 + \sum_{i \in W} \xi_i, \qquad \xi_i = \mathbf{1}\{a_i \geq \theta_i\}, \tag{3}$$

where C_0 and $\theta_i \in \{1, \dots, 9\}$ are functions of G. Indeed, for $i \in W$ the carry $c_{i+2} = \mathbf{1}\{a_{i+1} + t_{i+1} + c_{i+1} \geq 10\}$ equals $\mathbf{1}\{a_{i+1} + t_{i+1} \geq 10\}$, because $a_{i+1} + t_{i+1} \neq 9$; so the value of a_i influences no carry other than c_{i+1}. Proceeding upwards through the columns, every carry c_j with $j \notin \{i+1 : i \in W\}$ is therefore a function of G. In particular, for $i \in W$ the carry c_i into column i is a function of G (as $i-1$ is even, $i \neq i'+1$ for $i' \in W$), and $c_{i+1} = \mathbf{1}\{a_i \geq \theta_i\}$ with $\theta_i = 10 - t_i - c_i$. Since $t_i \in \{1, \dots, 8\}$, $\theta_i \in \{1, \dots, 9\}$, which proves the claim; truncating the count at column $k-1$ only changes C_0.

Given G, each ξ_i is Bernoulli with $\Pr(\xi_i = 1 \mid G) = \Pr(a_i \geq \theta_i) \in [\delta, 1-\delta]$, so $1 - \max_x \Pr(\xi_i = x \mid G) \geq \delta$. By (2), $\Pr(C = m \mid G) \leq K_0(\delta|W|)^{-1/2}$. By the multiplicative Chernoff bound [2, Thm. 4.5], $\Pr(|W| < \frac{9}{2}\delta|S(t)|) \leq e^{-9\delta|S(t)|/8}$. On the complementary event, $\Pr(C = m \mid G) \leq K_0(\frac{9}{2}\delta^2|S(t)|)^{-1/2}$, and the lemma follows. □

Lemma 9.2 (Sampling)**.** *Let $x_0, \dots, x_{D-1}$ be a uniformly random arrangement of a multiset of D digits with counts $N_0, \dots, N_9$, and let $y_0, y_1, \dots$ be independent with $\Pr(y_i = d) = N_d/D$. For $k \le D$, the laws of $(x_0, \dots, x_{k-1})$ and $(y_0, \dots, y_{k-1})$ are within total variation distance $k(k-1)/(2D)$.*

Proof. Realise both by drawing balls from an urn of D labelled balls, without and with replacement. The two procedures can be coupled to agree unless a ball is drawn twice with replacement, which has probability at most $\binom{k}{2}/D$ [1]. □

Lemma 9.3 (Low digits of the digit sum)**.** *Let a be uniform in $\{0, \dots, 10^D - 1\}$ and $t = s(a)$. There is K_2 such that for every modulus $M \ge 1$ the law of $t \bmod M$ is within total variation distance $K_2\, M/\sqrt{D}$ of the uniform law on $\{0, \dots, M-1\}$. (We use $M = 10^h$ and $M = 9 \cdot 10^h$.)*

Proof. The law of t is the D-fold convolution of the uniform law on $\{0, \dots, 9\}$. Convolutions of log-concave sequences are log-concave [6], so it is unimodal, and by (2) its largest point probability is $p_{\max} \le K/\sqrt{D}$. For a unimodal sequence, the sums over any two residue classes modulo M differ by at most $p_{\max}$, so each residue class has probability within $p_{\max}$ of $1/M$. Summing over the M classes gives the bound. □

Theorem 9.4 (Zero density of 2-cycles)**.** *There is an absolute constant K such that for all $D \ge 2$*

$$\frac{T(D)}{10^D} \;\le\; \frac{K}{\sqrt{\log D}}.$$

Proof. Let a be uniform in $\{0, \dots, 10^D - 1\}$, with digit counts $N = (N_d)$ and $t = s(a)$, and let E be the event that a is the odd point of a 2-cycle. By Theorem 6.2, $E \subseteq \{\mathrm{C}(a, t) = m(t)\}$ with $m(t) = s(t)/9$. Let ℓ be the number of digits of $9D$, so $t < 10^\ell$. Choose

$$h = \Big\lfloor \log_{10} \frac{\sqrt{D}}{\log D} \Big\rfloor, \qquad r = \lceil 4 \log_{10} D \rceil, \qquad k = \ell + r,$$

Typical multisets. Let $A = \{N_d \ge D/20 \text{ for all } d\}$. By the Chernoff bound, $\Pr(A^c) \le 10 e^{-cD}$.

Many switches. Let $U = \{|S(t) \cap [1, h)| \ge h/5\}$. Under the uniform law on $\{0, \dots, 10^h - 1\}$, the digits in the odd positions $1, 3, \dots$ below h are independent and uniform, each in $\{1, \dots, 8\}$ with probability $4/5$; their number is about $h/2$, so fewer than $h/5$ of them lie in $\{1, \dots, 8\}$ with probability at most e^{-ch}. By Lemma 9.3, $\Pr(U^c) \le e^{-ch} + K_2\, 10^h/\sqrt{D} \le e^{-ch} + K_2/\log D$.

Conditional bound. Fix $N \in A$ with $t = \sum_d d N_d$ satisfying U. Given N, the string a is a uniformly random arrangement of the multiset. Let $C^{(k)}$ count the carries of $a + t$ out of columns $0, \dots, k-1$. Columns $\ell, \dots, k-1$ have t-digit 0, so a carry passes column $k-1$ only if $a_\ell = \cdots = a_{k-1} = 9$; given $N \in A$ this has probability at most $(11/20 + k/D)^r \le (3/5)^r$ for large D. Hence $\Pr(E \mid N) \le \Pr(C^{(k)} = m(t) \mid N) + (3/5)^r$. By Lemma 9.2, replacing $(a_0, \dots, a_{k-1})$ by independent digits with law $N_d/D \ge 1/20$ changes this probability by at most $k(k-1)/(2D) \le k^2/(2D)$, and by Lemma 9.1 with $\delta = 1/20$ and $|S(t)| \ge h/5$,

$$\Pr(E \mid N) \le \frac{K_3}{\sqrt{h}} + e^{-ch} + \frac{k^2}{2D} + (3/5)^r.$$

Conclusion. Since $h \ge \frac{1}{2}\log_{10} D - \log_{10}\log D - 1$, $k = O(\log D)$ and $(3/5)^r \le D^{4\log_{10}(3/5)} \le D^{-0.88}$ (note $4\log_{10}(3/5) \approx -0.887$),

$$\frac{T(D)}{10^D} = \Pr(E) \le \Pr(A^c) + \Pr(U^c) + \sup_{N \in A \cap U} \Pr(E \mid N) = O\big((\log D)^{-1/2}\big). \qquad \square$$

Corollary 9.5. *The number of 2-cycles whose odd point is below N is $O\big(N/\sqrt{\log\log N}\big)$. In particular the proportion of multiples of 9 below N that lie on a 2-cycle tends to 0, so the number of 2-cycles does not grow linearly in N.*

Proof. Take $D = \lceil \log_{10} N \rceil$ in Theorem 9.4; every point of a 2-cycle is a multiple of 9 by Theorem 6.2. □

Remark 9.6 (Why the decay is so slow). The proof shows where the mechanism lives. The decision "exactly $s(t)/9$ carries" is made in the lowest $\approx \log_{10} t$ columns of $a + t$, and only the columns where t has a digit strictly between 0 and 9 act as independent switches. With $t \approx 4.5D$ there are only $O(\log D)$ such columns, so the carry count has standard deviation of order $\sqrt{\log D}$ and hits a prescribed value with probability of order $(\log D)^{-1/2}$. When t consists mostly of nines the switches disappear and the probability stays near 9/10 (Proposition 8.1); averaged over the spread of t, this produces the resonance peaks of Figure 6.

The same tools turn the resonance approximation into a theorem.

Theorem 9.7 (Resonance asymptotics)**.**

$$\frac{T(D)}{10^D} = \frac{1}{2} \sum_{t \equiv 9 \pmod{18}} P_D(t)\, \rho(t) \;+\; O\Big(\frac{(\log D)^2}{\sqrt{D}}\Big),$$

where $P_D(t)$ is the probability that D uniform random digits sum to t.

Proof. Condition on the digit counts N, which fixes t; if $t \not\equiv 9 \pmod{18}$ then E is empty, since $s(t)/9$ must be an integer and t odd. Otherwise $E = \{a_0 \text{ odd},\ \mathrm{C}(a,t) = m(t)\}$. With $k = \ell + r$ as above, replace C by $C^{(k)}$ at cost $(3/5)^r$, and replace the arrangement by independent digits with law N_d/D at cost $k^2/(2D)$ (Lemma 9.2). On the event $\max_d |N_d/D - \frac{1}{10}| \le \log D/\sqrt{D}$, whose complement has probability $O(D^{-2})$ by Hoeffding's inequality, the product of these k laws is within total variation $O(k \log D/\sqrt{D})$ of the product of uniform laws. Under uniform digits, $\Pr(a_0 \text{ odd},\ C^{(k)} = m(t)) = \frac{1}{2}\rho_k(t)$, where ρ_k is ρ computed with k columns, and $|\rho_k(t) - \rho(t)| \le 10^{-r}$. Averaging over N gives the main term with total error $O\big(k \log D/\sqrt{D} + k^2/D + (3/5)^r + 10^{-r}\big)$. Since $r = \lceil 4 \log_{10} D \rceil$ we have $(3/5)^r \le D^{-0.88} = o(D^{-1/2})$ and $10^{-r} \le D^{-4}$, and since $k = O(\log D)$ the total is $O((\log D)^2/\sqrt{D})$. □

Numerically the error in Theorem 9.7 is far smaller than this bound: at most $2 \cdot 10^{-6}$ for $100 \le D \le 200$ (Table 3).

10 Carry balance and the exact order of the 2-cycle density

The upper bound of Theorem 9.4 leaves open why 2-cycles are common at all. The answer is an exact identity: the number of carries that a 2-cycle demands is the *expected* number of carries.

Theorem 10.1 (Carry balance)**.** *Let $t \ge 0$, let K be at least the number of digits of t, and let $\mathrm{C}_K(a,t)$ count the carries out of columns $0, \dots, K-1$ in $a+t$. If a is uniform on $\{0, \dots, 10^K - 1\}$, then*

$$\mathbb{E}\ \mathrm{C}_K(a,t) = \frac{s(t) - t\,10^{-K}}{9}.$$

If instead a is uniform on the odd numbers in $\{0, \dots, 10^K - 1\}$ and t is odd with $t < 10^K$, then

$$\mathbb{E}\ \mathrm{C}_K(a,t) = \frac{s(t) + 1 - (t+1)\,10^{-K}}{9}.$$

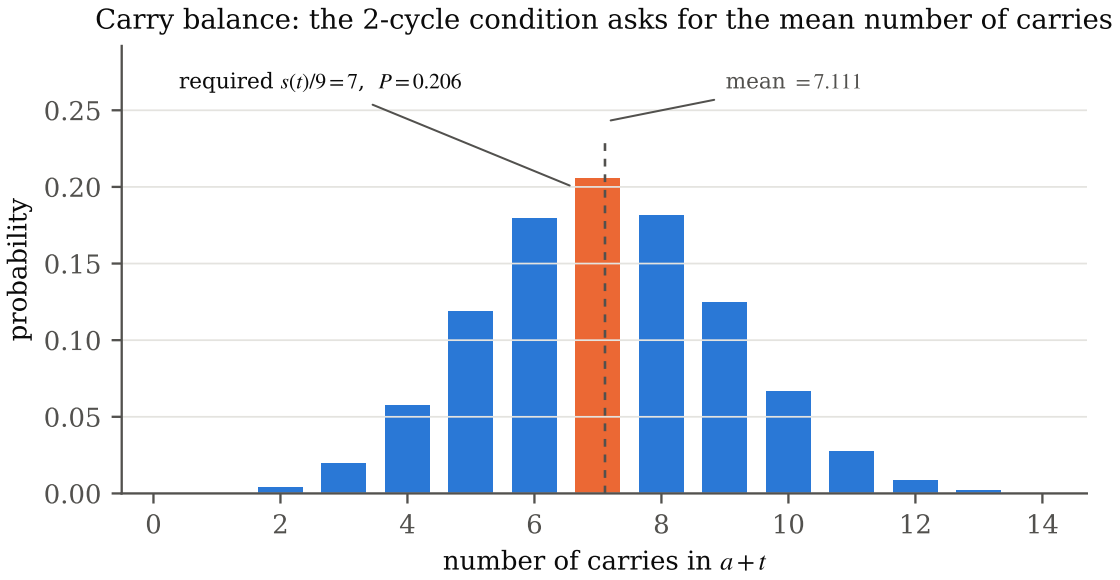


Figure 7: Exact distribution of the number of carries in $a + t$ for a typical $t = 4536271809123471$ ($s(t) = 63$), with a random and odd. The 2-cycle condition requires $7 = s(t)/9$ carries, next to the mean $64/9 \approx 7.111$ (Theorem 10.1); it holds with probability 0.206.

Proof. Let $b = (a + t) \bmod 10^K$ and $\omega = \mathbf{1}\{a + t \geq 10^K\}$, so $s(a + t) = s(b) + \omega$. Lemma 2.1 gives $9\,\mathrm{C}_K(a, t) = s(a) + s(t) - s(b) - \omega$. If a is uniform, so is b, hence $\mathbb{E}s(a) = \mathbb{E}s(b)$, and $\Pr(\omega = 1) = t/10^K$. If a is uniform on the $10^K/2$ odd numbers and t is odd, then b is uniform on the even numbers; their units digits average 5 and 4 respectively while the other digits are uniform, so $\mathbb{E}s(a) - \mathbb{E}s(b) = 1$. Overflow occurs iff $a \geq 10^K - t$; as $10^K - t$ and $10^K - 1$ are both odd, exactly $(t + 1)/2$ odd values of a qualify, so $\Pr(\omega = 1) = (t + 1)/10^K$. □

Both formulas were confirmed by exhaustive enumeration for $K = 4$ and several values of t.

Corollary 10.2 (A 2-cycle is a carry count near its mean)**.** *Let a be odd with $t = s(a) \equiv 9 \pmod{18}$. Then a lies on a 2-cycle if and only if $\mathrm{C}(a, t) = s(t)/9$. For every $K > \log_{10} t$, this required value differs from the mean $\mathbb{E}\,\mathrm{C}_K(a', t)$ over uniformly random odd $a' < 10^K$ by $\big(1 - (t + 1)10^{-K}\big)/9 \in [0, 1/9)$; the difference is 0 exactly when $t + 1 = 10^K$.*

This is the conceptual reason for everything in Sections 8–9: the carry count is a sum of about $\log_{10} t$ weakly dependent terms, so it equals its mean with probability of order $(\log t)^{-1/2}$ (Figure 7), except when the digits of t are almost all 9, where the count is nearly deterministic and the probability is close to $9/10$. We now turn this into a sharp theorem.

Lemma 10.3 (Variance of carries)**.** *Let $t \geq 0$ and let $a_0, a_1, \ldots, a_{k-1}$ be independent digits, where a_0 may have any law and $\Pr(a_i = d) \geq \delta$ for all $d \in \{0, \ldots, 9\}$ and $1 \leq i \leq k - 1$. Let c_j be the carry out of column $j - 1$ ($1 \leq j \leq k$), so that $\mathrm{C}_k(a, t) = \sum_{j=1}^{k} c_j$. Then*

$$|\operatorname{Cov}(c_i, c_j)| \leq (1 - 9\delta)^{j-i} \quad (1 \leq i < j \leq k), \qquad \operatorname{Var} \mathrm{C}_k(a, t) \leq \Big(1 + \frac{2}{9\delta}\Big)k.$$

Proof. Fix $1 \leq i < j \leq k$. The carry c_j is a function of c_i and of the digits in columns $i, \ldots, j - 1$, all of which are at least 1 and independent of c_i. Run the carry recursion through these columns

twice with the same digits, once from carry-in 0 and once from carry-in 1. In column l the two runs produce the same carry-out unless $a_l + t_l = 9$, an event of probability $\Pr(a_l = 9 - t_l) \le 1 - 9\delta$; once the runs agree they agree forever. Hence, with $\Delta = \Pr(c_j = 1 \mid c_i = 1) - \Pr(c_j = 1 \mid c_i = 0)$, $0 \le \Delta \le (1-9\delta)^{j-i}$ by independence of the columns. Since $\mathrm{Cov}(c_i, c_j) = \mathrm{Var}(c_i)\,\Delta$ and $\mathrm{Var}(c_i) \le \frac{1}{4}$, the covariance bound follows (with room to spare). Note that a_0 is never used: column 0 only influences c_1, and the bound for pairs involving c_1 uses columns $1, \dots, j-1$. Finally

$$\mathrm{Var}\,\mathrm{C}_k \le \sum_{j=1}^{k} \mathrm{Var}(c_j) + 2\sum_{i<j} (1-9\delta)^{j-i} \le \frac{k}{4} + 2k \sum_{d \ge 1} (1-9\delta)^d \le \Big(1 + \frac{2}{9\delta}\Big) k. \qquad \square$$

Lemma 10.4 (Local lower bound for switch sums)**.** *Let $\xi_1, \dots, \xi_n$ be independent Bernoulli variables with success probabilities $p_i \in [\delta, 1-\delta]$, and let $S = \sum_i \xi_i$, $\mu = \sum_i p_i$, $V = \sum_i p_i(1-p_i)$. For every $A > 0$ there are constants $c_{A,\delta} > 0$ and $n_{A,\delta}$, depending only on A and δ, such that for all $n \ge n_{A,\delta}$, all such $p_1, \dots, p_n$ and every integer x with $|x - \mu| \le A\sqrt{n}$,*

$$\Pr(S = x) \ge \frac{c_{A,\delta}}{\sqrt{n}}.$$

Proof. Note $\delta(1-\delta)n \le V \le n/4$.

(a) Unimodality. The generating polynomial $\prod_i (1 - p_i + p_i z)$ has only real roots, so its coefficients $\Pr(S = j)$ form a log-concave, hence unimodal, sequence [6].

(b) Normal approximation. By the Berry–Esseen inequality for independent, non-identically distributed summands [7], $\sup_y |\Pr(S \le y) - \Phi((y-\mu)/\sqrt{V})| \le C_0 \sum_i \mathbb{E}|\xi_i - p_i|^3 / V^{3/2} \le C_0/\sqrt{V}$, since $\mathbb{E}|\xi_i - p_i|^3 \le p_i(1-p_i)$.

(c) Mass on each side. Let $w = \lceil \sqrt{V} \rceil$, $J_- = \{x - w, \dots, x - 1\}$ and $J_+ = \{x + 1, \dots, x + w\}$. In standard units $|x - \mu|/\sqrt{V} \le A' := A/\sqrt{\delta(1-\delta)}$, and each of $J_\pm$ corresponds to an interval of length at least $1 - 1/\sqrt{V}$ within distance $A' + 2$ of 0. By (b), $\Pr(S \in J_\pm) \ge \eta - 2C_0/\sqrt{V}$ with $\eta = \frac{1}{2}\varphi(A' + 2)$, where φ is the standard normal density; this is at least $\eta/2$ once n is large in terms of A, δ.

(d) Point mass. Let M be a mode of S. If $M \ge x$, unimodality gives $\Pr(S = y) \le \Pr(S = x)$ for all $y \in J_-$, so $\eta/2 \le \Pr(S \in J_-) \le w \Pr(S = x)$. If $M < x$, the same argument applies to J_+. In both cases $\Pr(S = x) \ge \eta/(2w) \ge \eta/(4\sqrt{V}) \ge \eta/(2\sqrt{n})$. $\square$

Theorem 10.5 (Exact order of the 2-cycle density)**.** *There are absolute constants $c, K > 0$ and D_0 such that for all $D \ge D_0$*

$$\frac{c}{\sqrt{\log D}} \;\le\; \frac{T(D)}{10^D} \;\le\; \frac{K}{\sqrt{\log D}}.$$

Equivalently, the number of 2-cycles with odd point below N is of exact order $N/\sqrt{\log\log N}$.

Plan of the lower bound. The proof conditions in three nested layers. Each layer fixes part of the randomness and keeps the rest independent, and Table 5 lists the objects involved.

1. *Digit counts.* Fixing the multiset N of digits of a fixes $t = s(a)$ and the target $m = s(t)/9$. Given N, the arrangement of the digits is uniform. By Lemma 9.2, its lowest $k = O(\log D)$ digits may be replaced by *independent* digits with law $p_d = N_d/D$, at a cost of $k^2/(2D)$.

2. *Non-switch digits.* Under the independent law we condition on G, which consists of all digits except those at the active switches W. The set W is a function of G: whether $i \in S(t)$ is active depends only on a_{i+1}, and $i + 1$ is even, so it is never itself a switch. The remaining digits

a_i, $i \in W$, are independent of G and of each other, and each influences only the carry out of its own column. Therefore $\mathrm{C}_k = C_0 + \sum_{i \in W} \xi_i$ with C_0 determined by G and independent Bernoulli summands ξ_i.

3. *Good conditionings.* Two events determined by G make the local estimate applicable. B_1 says that the conditional mean $C_0 + \mu_W$ is within $2R\sqrt{k}$ of m. It has probability at least $\frac{1}{2}$ because the unconditional carry count is concentrated near m (carry balance and the variance lemma) and the Bernoulli part is concentrated near its mean. B_2 says that $|W|$ is of order h; it fails with probability $e^{-\Omega(h)}$ by the Chernoff bound. On $B_1 \cap B_2$ the target $x = m - C_0$ of the Bernoulli sum lies within $A\sqrt{|W|}$ of its mean, with A bounded because $k = O(h)$. Lemma 10.4 then gives probability of order $1/\sqrt{|W|} \asymp 1/\sqrt{\log D}$.

object	determined by	role
N, t, m	digit multiset	fixes the target; arrangement uniform given N
A', U'	N	typical counts; $t \equiv 9 \pmod{18}$ with $\geq h/5$ switch positions
p	N	independent-digit approximation of the low digits
$W \subseteq S(t)$	digits at even positions (part of G)	active switches: $a_{i+1} + t_{i+1} \neq 9$
G	all digits outside W (incl. a_0)	fixes C_0 and the thresholds θ_i
ξ_i, $i \in W$	a_i	independent Bernoulli given G, parameters in $[\delta, 1-\delta]$
B_1, B_2	G	conditional mean near m; many switches
$x = m - C_0$	G	target for $\sum \xi_i$, within the local-limit window

Table 5: The objects in the proof of the lower bound in Theorem 10.5.

Proof. The upper bound is Theorem 9.4. For the lower bound we keep the notation of its proof: ℓ is the number of digits of $9D$, $h = \lfloor \log_{10}(\sqrt{D}/\log D) \rfloor$, $r = \lceil 4 \log_{10} D \rceil$, $k = \ell + r$ and $\delta = 1/20$. For large D,

$$\tfrac{1}{4} \log_{10} D \leq h, \qquad k \leq 6 \log_{10} D. \tag{4}$$

All constants below depend only on δ and on absolute constants, never on D.

Step 1: typical digit counts and digit sums. Let A' be the event $\max_d |N_d/D - \frac{1}{10}| \leq \log D/\sqrt{D}$; by Hoeffding's inequality $\Pr(A'^c) = O(D^{-2})$. Let U' be the event that $t \equiv 9 \pmod{18}$ and $|S(t) \cap [1, h)| \geq h/5$. Both conditions depend only on $t \bmod 9 \cdot 10^h$. Under the uniform law on $\mathbb{Z}/9 \cdot 10^h \cong \mathbb{Z}/9 \times \mathbb{Z}/10^h$, the residue mod 9 is uniform and independent of the last h digits, so $t \equiv 0 \pmod 9$ has probability $1/9$; the units digit is odd with probability $1/2$; and, independently of the units digit, at least $h/5$ of the roughly $h/2$ odd positions below h carry a digit in $\{1, \ldots, 8\}$ with probability at least $1 - e^{-c_1 h}$. By Lemma 9.3 with $M = 9 \cdot 10^h$,

$$\Pr(U') \geq \tfrac{1}{18}\big(1 - e^{-c_1 h}\big) - K_2 \frac{9 \cdot 10^h}{\sqrt{D}} \geq \tfrac{1}{20} \quad \text{for large } D.$$

Step 2: reduction to independent digits. Fix digit counts N in A' such that $t = t(N)$ satisfies U', and let $m = s(t)/9$. Then $p_d = N_d/D \geq 1/20$ for all d. Given N, a is a uniform arrangement,

and $E \supseteq \{a_0 \text{ odd},\ \mathrm{C}_k(a,t) = m\} \setminus F$, where F is the event that a carry passes column $k-1$; as in Theorem 9.4, $\Pr(F \mid N) \le (3/5)^r$. By Lemma 9.2,

$$\Pr(E \mid N) \ge \Pr_p\big(a_0 \text{ odd},\ \mathrm{C}_k = m\big) - \frac{k^2}{2D} - (3/5)^r, \tag{5}$$

where $\Pr_p$ refers to independent digits with law p. We bound the first term from below by $\Pr_p(a_0 \text{ odd}) \cdot \Pr_p(\mathrm{C}_k = m \mid a_0 \text{ odd})$ with $\Pr_p(a_0 \text{ odd}) \ge \frac{1}{2} - 5\log D/\sqrt{D} \ge \frac{1}{4}$.

Step 3: the carry count is concentrated near m. Let $X = \mathrm{C}_k$ under $\Pr_p(\cdot \mid a_0 \text{ odd})$. Under uniform digits with a_0 odd, Theorem 10.1 gives $\mathbb{E}X = m + \big(1-(t+1)10^{-k}\big)/9$. The laws of $(a_0, \dots, a_{k-1})$ under p and under the uniform law (both conditioned on a_0 odd) differ in total variation by at most $10k\log D/\sqrt{D}$, and $0 \le X \le k$, so the mean changes by at most $10k^2 \log D/\sqrt{D} = o(1)$. Hence $|\mathbb{E}X - m| \le 1$ for large D. By Lemma 10.3, which allows column 0 to have an arbitrary law (here: uniform on the odd digits) and requires $\Pr(a_i = d) \ge \delta$ only for $1 \le i \le k-1$, we have $\operatorname{Var} X \le K_3 k$ with $K_3 = 1 + 2/(9\delta)$. Put $R = \max(1, 4\sqrt{K_3})$. Chebyshev's inequality gives, for $k \ge 4$,

$$\Pr\big(|X - m| > R\sqrt{k}\big) \le \Pr\big(|X - \mathbb{E}X| > R\sqrt{k} - 1\big) \le \frac{K_3 k}{(R\sqrt{k}/2)^2} \le \frac{1}{4}.$$

Step 4: switches. Condition on G, the digits outside the switch set W of Lemma 9.1 (with a_0 included in G). Then $X = C_0 + \sum_{i \in W} \xi_i$, where C_0 is a function of G and, given G, the ξ_i are independent Bernoulli variables with success probabilities $p_i \in [\delta, 1-\delta]$. Write $\mu_W = \sum_{i \in W} p_i$ and $Y = \sum_{i \in W} \xi_i - \mu_W$. Since $\operatorname{Var}(Y \mid G) \le |W|/4 \le k/4$, Chebyshev's inequality gives $\Pr(|Y| > R\sqrt{k}) \le 1/(4R^2) \le 1/4$. As $C_0 + \mu_W - m = (X - m) - Y$, the event

$$B_1 = \big\{|C_0 + \mu_W - m| \le 2R\sqrt{k}\big\},$$

which is determined by G, satisfies $\Pr(B_1) \ge 1 - \frac{1}{4} - \frac{1}{4} = \frac{1}{2}$. Let $B_2 = \{|W| \ge w_0\}$ with $w_0 = \frac{9}{2}\delta \cdot \frac{h}{5}$; also determined by G. Since $|S(t)| \ge h/5$, the Chernoff bound in Lemma 9.1 gives $\Pr(B_2^c) \le e^{-9\delta h/40}$. Hence

$$\Pr(B_1 \cap B_2) \ge \tfrac{1}{2} - e^{-9\delta h/40} \ge \tfrac{1}{4} \quad \text{for large } D.$$

Step 5: local lower bound. On $B_1 \cap B_2$, the target value $x = m - C_0$ satisfies $|x - \mu_W| \le 2R\sqrt{k} \le A\sqrt{|W|}$, where $A = 2R\sqrt{k/w_0}$ is bounded by an absolute constant because of (4). Since $|W| \ge w_0 \to \infty$, Lemma 10.4, applied conditionally on G, gives

$$\Pr(X = m \mid G) = \Pr\Big(\sum_{i \in W} \xi_i = x \;\Big|\; G\Big) \ge \frac{c_{A,\delta}}{\sqrt{|W|}} \ge \frac{c_{A,\delta}}{\sqrt{k}}$$

uniformly on $B_1 \cap B_2$. Therefore $\Pr_p(\mathrm{C}_k = m \mid a_0 \text{ odd}) \ge \frac{1}{4}\, c_{A,\delta}/\sqrt{k} \ge c_4/\sqrt{\log D}$ by (4).

Step 6: conclusion. Combining with (5) and Step 2, for every admissible N, $\Pr(E \mid N) \ge \frac{1}{4}c_4/\sqrt{\log D} - k^2/(2D) - (3/5)^r \ge c_5/\sqrt{\log D}$ for large D. Averaging over N,

$$\frac{T(D)}{10^D} = \Pr(E) \ge \Pr(A' \cap U') \cdot \frac{c_5}{\sqrt{\log D}} \ge \Big(\tfrac{1}{20} - O(D^{-2})\Big)\frac{c_5}{\sqrt{\log D}} \ge \frac{c}{\sqrt{\log D}}. \qquad \square$$

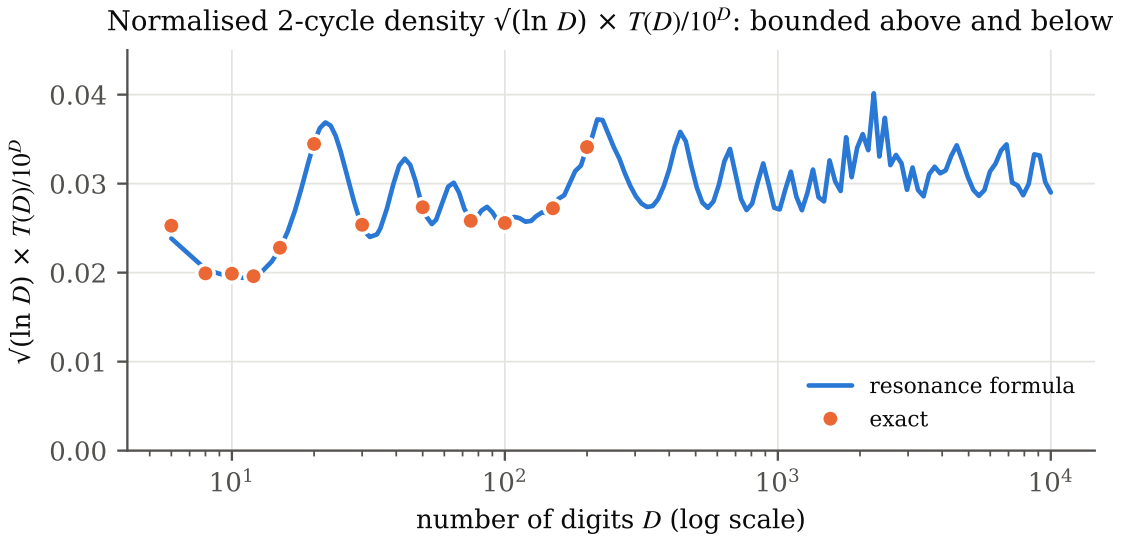


Figure 8: The 2-cycle density multiplied by $\sqrt{\ln D}$ stays between positive bounds, as Theorem 10.5 asserts; the log-periodic modulation of Figure 6 remains.

10.1 The oscillation law

We now determine the normalised density exactly, up to a factor $1 + o(1)$. This proves that it oscillates, identifies its limit points and settles Conjecture 8.2 (in a corrected form: the limiting ratio is $\sqrt{301}/11 \approx 1.577$, not 1.4). Three new ingredients are needed: an exact formula for the variance of the carry count, a sharp extremal bound for the digit averages that this formula involves, and a local limit theorem with the sharp constant.

Throughout this subsection a has independent digits, a_0 uniform on $\{1, 3, 5, 7, 9\}$ and a_i uniform on $\{0, \dots, 9\}$ for $i \geq 1$. For $t \geq 0$ let c_i ($i \geq 1$) be the carry into column i of $a+t$, let $q_i = \Pr(c_i = 1)$, and let $C = C(a, t) = \sum_{i \geq 1} c_i$. Then $\rho(t) = \Pr(C = s(t)/9)$. Write $\{y\}$ for the fractional part of a real number y and

$$\varphi(x) = x(1 - x).$$

Lemma 10.6 (Exact carry variance)**.** *For every $t \geq 0$ and $1 \leq i \leq j$,*

$$q_{i+1} = \frac{t_i + q_i}{10}, \qquad \operatorname{Cov}(c_i, c_j) = 10^{-(j-i)}\, q_i(1 - q_i), \qquad \operatorname{Var} C = \frac{11}{9} \sum_{i \geq 1} \varphi(q_i).$$

Moreover $|q_i - \{t/10^i\}| \leq 10^{1-i}$ for all $i \geq 1$.

Proof. Given $c_i = c$, column i carries out iff $a_i \geq 10 - t_i - c$, which has probability $(t_i + c)/10$ (also when $t_i + c = 10$). So $\mathbb{E}[c_{i+1} \mid c_1, \dots, c_i] = (t_i + c_i)/10$. Taking expectations gives the recursion for q_i. Subtracting it gives $\mathbb{E}[c_{i+1} - q_{i+1} \mid c_1, \dots, c_i] = (c_i - q_i)/10$, and by induction $\mathbb{E}[c_j - q_j \mid c_1, \dots, c_i] = 10^{-(j-i)}(c_i - q_i)$. Multiplying by $c_i - q_i$ and taking expectations gives the covariance. Above the top digit of t we have $t_i = 0$ and $q_{i+1} = q_i/10$, so $\sum_i q_i < \infty$ and all the series below converge absolutely. Hence

$$\operatorname{Var} C = \sum_i \varphi(q_i)\Big(1 + 2\sum_{d \geq 1} 10^{-d}\Big) = \frac{11}{9} \sum_i \varphi(q_i).$$

Finally $x_i = \{t/10^i\} = (t \bmod 10^i)/10^i$ satisfies $x_1 = t_0/10$ and $x_{i+1} = (t_i + x_i)/10$, so $q_i - x_i = 10^{1-i}(q_1 - x_1)$, and $|q_1 - x_1| \le 1$. □

So the variance of the carry count is an additive functional of the fractional parts $\{t/10^i\}$. A digit 9 or 0 in t pushes the next fractional part towards 1 or 0, where φ vanishes. This is the exact form of the resonance mechanism.

Lemma 10.7 (Extremal digit averages)**.** *Let $x_1 \in [0,1]$, let $d_1, \dots, d_n \in \{0, \dots, 9\}$ and $x_{i+1} = (d_i + x_i)/10$. Then*

$$\sum_{i=1}^{n} \varphi(x_i) \le \frac{30}{121}\, n + \frac{5}{22}.$$

The constant $30/121$ is best possible: for the digits $4, 5, 4, 5, \dots$ the points x_i converge to the 2-cycle $\{5/11, 6/11\}$, where $\varphi = 30/121$.

Proof. Put $u_i = x_i - \frac{1}{2} \in [-\frac{1}{2}, \frac{1}{2}]$ and $e_i = d_i - \frac{9}{2}$, so that $u_{i+1} = (e_i + u_i)/10$ and $|e_i| \ge \frac{1}{2}$. For $\alpha = \frac{1}{11}$, $\beta = \frac{1}{110}$ and all real u and e with $|e| \ge \frac{1}{2}$, minimising over u gives

$$\alpha u^2 + \beta(u+e)^2 \;\ge\; \frac{\alpha\beta}{\alpha+\beta}\, e^2 = \frac{e^2}{121} \;\ge\; \frac{1}{484}.$$

With $u = u_i$, $e = e_i$ and $\beta(u_i + e_i)^2 = \frac{10}{11} u_{i+1}^2$ this reads $u_i^2 - \frac{1}{484} \ge \frac{10}{11}\big(u_i^2 - u_{i+1}^2\big)$. Summing, $\sum_{i=1}^{n} u_i^2 \ge \frac{n}{484} - \frac{10}{11} \cdot \frac{1}{4}$. Since $\varphi(x_i) = \frac{1}{4} - u_i^2$, the bound follows. The 2-cycle: the composition $x \mapsto (4 + (5+x)/10)/10$ fixes $5/11$, and $(5 + 5/11)/10 = 6/11$. □

Lemma 10.8 (Smoothness of Bernoulli sums)**.** *Let S be a sum of $n \ge 1$ independent Bernoulli variables with success probabilities in $[\delta, 1-\delta]$. Then for every integer x*

$$|\Pr(S = x+1) - \Pr(S = x)| \le \frac{\pi}{4\delta(1-\delta)\, n}.$$

Proof. Let $\psi(\theta) = \mathbb{E} e^{i\theta S}$. For each summand, $|1 - p + pe^{i\theta}|^2 = 1 - 2p(1-p)(1-\cos\theta) \le \exp(-4\delta(1-\delta)\theta^2/\pi^2)$ for $|\theta| \le \pi$, using $1 - \cos\theta \ge 2\theta^2/\pi^2$. So $|\psi(\theta)| \le e^{-cn\theta^2}$ with $c = 2\delta(1-\delta)/\pi^2$. By Fourier inversion,

$$|\Pr(S = x+1) - \Pr(S = x)| = \Big|\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-ix\theta}(e^{-i\theta} - 1)\psi(\theta)\, d\theta\Big| \le \frac{1}{2\pi}\int_{-\infty}^{\infty} |\theta| e^{-cn\theta^2} d\theta = \frac{1}{2\pi cn}.$$ □

Theorem 10.9 (Sharp local limit theorem for carries)**.** *For every $\lambda \in (0, 1]$ and $\varepsilon > 0$ there is n_0 with the following property. Let t have ℓ digits and suppose its switch set $S(t)$ (Lemma 9.1) has $|S(t)| \ge \max(n_0, \lambda\ell)$. Then $V = \operatorname{Var} C$ and $\mu = \mathbb{E} C$ satisfy, for every integer m with $|m - \mu| \le 1$,*

$$\Big|\sqrt{2\pi V}\, \Pr(C = m) - 1\Big| \le \varepsilon.$$

Proof. Let $p(x) = \Pr(C = x)$ and $\delta = \frac{1}{10}$.

Size of V. Lemma 9.1 (whose proof applies to the full carry count) gives $C = C_0 + \sum_{i \in W} \xi_i$, where, given the digits G outside W, the ξ_i are independent Bernoulli variables with parameters in $[\delta, 1-\delta]$, and $\Pr(i \in W) \ge 9\delta$ independently over $i \in S(t)$. Hence $V \ge \mathbb{E} \operatorname{Var}(C \mid G) \ge \delta(1-\delta)\, \mathbb{E}|W| \ge 0.08\, |S(t)|$. In the other direction, Lemma 10.6 gives $V \le \frac{11}{9}(\frac{\ell}{4} + \sum_{i > \ell} q_i) \le \ell/3 + 1 \le (1/\lambda + 1)|S(t)|$, because $q_i \le 10^{\ell - i}$ for $i > \ell$.

Smoothness. Given G, $p(\cdot \mid G)$ is a shift of the law of a Bernoulli sum with $|W|$ terms, so Lemma 10.8 and the Chernoff bound $\Pr(|W| < \frac{9}{2}\delta|S(t)|) \le e^{-9\delta|S(t)|/8}$ give

$$\sup_x |p(x+1) - p(x)| \le L := \frac{K}{|S(t)|} + e^{-|S(t)|/90}$$

for an absolute constant K.

Central limit theorem. The carries $c_1, c_2, \dots$ form a non-homogeneous Markov chain on $\{0,1\}$ whose i-th transition matrix has rows $(1 - \frac{t_i}{10}, \frac{t_i}{10})$ and $(1 - \frac{t_i+1}{10}, \frac{t_i+1}{10})$. Its Dobrushin contraction coefficient is therefore $\frac{1}{10}$ for every i and every t. Truncating C after column $\ell + r$ changes it with probability at most $\sum_{i>\ell+r} q_i \le 10^{-r}$. Dobrushin's central limit theorem for non-homogeneous Markov chains [4, 5], applied to bounded functionals with contraction coefficients bounded away from 1, shows that $(C - \mu)/\sqrt{V}$ converges in law to the standard normal along every sequence of t with $V \to \infty$. By the usual compactness argument, for every $\varepsilon' > 0$ there is V_0 such that $\sup_y |\Pr(C \le \mu + y\sqrt{V}) - \Phi(y)| \le \varepsilon'$ whenever $V \ge V_0$.

Combination. Fix $\eta \in (0,1)$ and let $J = \{x \in \mathbb{Z} : |x - m| \le \eta\sqrt{V}\}$, so $\big||J| - 2\eta\sqrt{V}\big| \le 1$. For $x \in J$, $|p(x) - p(m)| \le L\eta\sqrt{V}$, so $\big|\sum_{x\in J} p(x) - |J|\,p(m)\big| \le |J|\,L\eta\sqrt{V}$. Since $|m - \mu| \le 1$, the central limit theorem gives $\sum_{x\in J} p(x) = 2\eta\,\phi(0)\big(1 + O(\eta^2 + V^{-1/2})\big) + O(\varepsilon')$, with $\phi(0) = 1/\sqrt{2\pi}$. Dividing by $|J|$,

$$\sqrt{V}\,p(m) = \phi(0)\big(1 + O(\eta^2 + V^{-1/2})\big) + O\Big(\frac{\varepsilon'}{\eta} + \frac{1}{\eta\sqrt{V}}\Big) + O(L\eta V).$$

By the bounds on V, $L\eta V \le K'\eta/\lambda$ once $|S(t)|$ is large. Choose η so small that the η-terms are below $\varepsilon/3$, then $\varepsilon' = \varepsilon\eta/3$, and then n_0 so large that $V \ge 0.08 n_0 \ge V_0$ and the remaining terms are below $\varepsilon/3$. □

Theorem 10.10 (Oscillation law)**.** *Let $h = h_D = \lfloor \log_{10}(\sqrt{D}/\log D) \rfloor$ and*

$$B_D = \sum_{i>h} \varphi\big(\{4.5D/10^i\}\big).$$

Then, as $D \to \infty$,

$$\frac{T(D)}{10^D} = \frac{1 + o(1)}{36\sqrt{\frac{22\pi}{9}\big(\frac{h}{6} + B_D\big)}}.$$

Proof. By Theorem 9.7 it suffices to evaluate $\frac{1}{2}\sum_{t\equiv 9\,(18)} P_D(t)\rho(t)$ up to a factor $1 + o(1)$; the error there is $O((\log D)^2/\sqrt{D})$, while the claimed main term is of order $(\log D)^{-1/2}$. Let $\varepsilon_D = (\log D)^{-1/8}$ and $x_i(y) = \{y/10^i\}$. Call $t \equiv 9 \pmod{18}$ *good* if

(G1) $|t - 4.5D| \le \sqrt{D}\log D$;

(G2) $|S(t) \cap [1, h)| \ge h/5$;

(G3) $\big|\sum_{i=1}^{h} \varphi(x_i(t)) - h/6\big| \le \varepsilon_D h$.

Bad t are negligible. (G1) fails with probability $e^{-c\log^2 D}$. Conditions (G2), (G3) and $t \equiv 9 \pmod{18}$ depend only on $t \bmod 9 \cdot 10^h$, whose law is within total variation $9K_2\,10^h/\sqrt{D} \le 9K_2/\log D$ of uniform (Lemma 9.3). Under the uniform law, $t \equiv 0 \pmod 9$ and t odd have probability $\frac{1}{18}$ jointly, and independently of them the digits $t_1, \dots, t_{h-1}$ are uniform and independent. Then (G2) fails with probability e^{-ch}, as in Theorem 9.4. For (G3), each x_i is uniform on the grid

$10^{-i}\mathbb{Z}\cap[0,1)$, so $\mathbb{E}\varphi(x_i)=\frac16-\frac16 10^{-2i}$. Also $\varphi(x_j)$ differs by at most $10^{-(j-i)}$ from a function of $t_i,\dots,t_{j-1}$, which is independent of x_i; so $|\operatorname{Cov}(\varphi(x_i),\varphi(x_j))|\le 2\cdot 10^{-(j-i)}$ and the variance of the sum is $O(h)$. By Chebyshev's inequality (G3) fails with probability $O(1/(\varepsilon_D^2 h))=O((\log D)^{-3/4})$. Since $\rho\le 1$, bad t contribute $o((\log D)^{-1/2})$.

Variance of good t. By Lemma 10.6, and since φ is 1-Lipschitz, $V(t)=\frac{11}{9}\big(\sum_{i\ge1}\varphi(x_i(t))+O(1)\big)$. The terms $i\le h$ contribute $h/6+O(\varepsilon_D h)$ by (G3). Let $\omega=\lceil 3\log_{10}\log D\rceil+2$. The terms $h<i\le h+\omega$ of both $\sum_i\varphi(x_i(t))$ and B_D are at most $\omega/4$ in total. For $i>h+\omega$, φ is 1-Lipschitz on $\mathbb{R}/\mathbb{Z}$ (since $\varphi(0)=\varphi(1)$), so $|\varphi(x_i(t))-\varphi(x_i(4.5D))|\le|t-4.5D|/10^i$. By (G1) and $10^h\ge\sqrt{D}/(10\log D)$, these differences sum to at most $20\log^2 D/10^\omega\le 1$. Hence

$$V(t)=\frac{11}{9}\Big(\frac{h}{6}+B_D\Big)\big(1+o(1)\big)$$

uniformly over good t, because $h/6+B_D\ge h/6$ and the errors are $O(\varepsilon_D h+\log\log D)=o(h)$.

Local limit. A good t has $\ell\le\log_{10}D+2\le 3h$ digits for large D and $|S(t)|\ge h/5\ge\ell/15$. By Theorem 10.1 (letting $K\to\infty$) $\mu=\mathbb{E}C=(s(t)+1)/9$, so the target $m=s(t)/9$ satisfies $|m-\mu|=\frac19$. Theorem 10.9 with $\lambda=\frac{1}{15}$ gives $\rho(t)=(1+o(1))(2\pi V(t))^{-1/2}$ uniformly over good t.

Conclusion. The good t carry $\frac12 P_D(\text{good})=\frac{1}{36}+o(1)$, and on them $\rho(t)=(1+o(1))\big(\frac{22\pi}{9}(\frac{h}{6}+B_D)\big)^{-1/2}$. □

Corollary 10.11 (The normalised density oscillates)**.** *Let $\Phi(D)=\sqrt{\log_{10}D}\,T(D)/10^D$ and $K_0=1/(12\sqrt{11\pi})$. Then*

$$\limsup_{D\to\infty}\Phi(D)=K_0\sqrt{6}\approx 0.03472,\qquad \liminf_{D\to\infty}\Phi(D)=K_0\sqrt{\tfrac{726}{301}}\approx 0.02201,$$

and the set of limit points of $\Phi(D)$ is the whole interval between them. Moreover $\Phi(D)\to K_0\sqrt{3}\approx 0.02455$ for almost all D: for every $\varepsilon>0$, the proportion of $D\in[10^j,10^{j+1})$ with $|\Phi(D)-K_0\sqrt{3}|>\varepsilon$ tends to 0 as $j\to\infty$. In particular $\Phi(D)$ has no limit, and

$$\frac{\limsup\Phi}{\liminf\Phi}=\frac{\sqrt{301}}{11}=1.5772\ldots$$

The lim sup *is approached along $D=\lfloor 10^j/4.5\rfloor$, where the top digits of the typical digit sum are nines. The* lim inf *is approached along $D=\lfloor\frac29 y_j\rfloor$, where $y_j=4545\cdots$ has j digits. The same holds with $\sqrt{\ln D}$ in place of $\sqrt{\log_{10}D}$, all values multiplied by $\sqrt{\ln 10}$.*

Proof. Let ℓ_D be the number of digits of $\lfloor 4.5D\rfloor$. Then $h/\log_{10}D\to\frac12$ and $(\ell_D-h)/\log_{10}D\to\frac12$. Positions $i>\ell_D$ contribute at most $\sum_{i>\ell_D}10^{\ell_D-i}\le\frac19$ to B_D, so Lemma 10.7 gives $0\le B_D\le\frac{30}{121}(\ell_D-h)+1$. Write $B_D=\frac12\log_{10}D\cdot\beta_D$. Then $0\le\beta_D\le\frac{30}{121}+o(1)$, and Theorem 10.10 becomes

$$\Phi(D)=\frac{1+o(1)}{36\sqrt{11\pi/9}}\Big(\frac16+\beta_D\Big)^{-1/2}=K_0\Big(\frac16+\beta_D\Big)^{-1/2}+o(1),$$

with $\frac16+\frac{30}{121}=\frac{301}{726}$. For $D=\lfloor 10^j/4.5\rfloor$ we have $10^j-4.5<4.5D\le 10^j$. So $\varphi(\{4.5D/10^i\})\le 4.5\cdot10^{-i}$ for $i\le j$ and $\le 10^{j-i}$ for $i>j$, giving $B_D=O(1)$ and $\beta_D\to0$. For $D=\lfloor\frac29 y_j\rfloor$, $\{4.5D/10^i\}$ lies within $6\cdot10^{-i}$ of $5/11$ or $6/11$ for $h<i\le j$. So $B_D\ge\frac{30}{121}(j-h)-O(1)$ and $\beta_D\to\frac{30}{121}$. Finally, $|B_{D+1}-B_D|\le\sum_{i>h}4.5\cdot10^{-i}\to0$ because φ is 1-Lipschitz on the circle. Hence $\Phi(D+1)-\Phi(D)\to0$, and a sequence with vanishing increments that oscillates between two values has every intermediate value as a limit point.

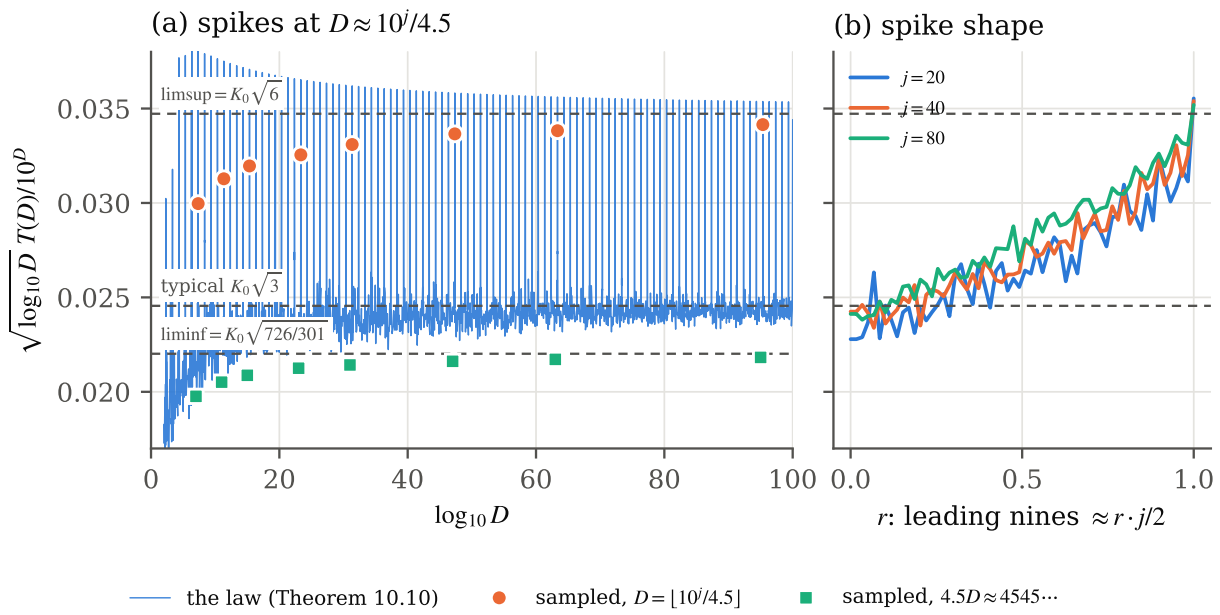


Figure 9: The oscillation law. (a) The main term of Theorem 10.10, evaluated densely near the resonances $D \approx 10^j/4.5$ and on a uniform grid in $\log_{10} D$. For most D it is close to the typical value $K_0\sqrt{3}$. Near $D = 10^j/4.5$ it spikes towards $K_0\sqrt{6}$, and near $4.5D \approx 4545\cdots$ it dips towards $K_0\sqrt{726/301}$. The finite-D formula overshoots $K_0\sqrt{6}$ for moderate D because $h/\log_{10} D \to \frac{1}{2}$ only logarithmically. Circles and squares are sampled values of the exact main term, $\frac{1}{2}\sum_t P_D(t)\rho(t)$, at peaks and troughs, which approach the proved limits. (b) Spike shape: the law at $D = (10^j/4.5)(1-10^{-rj/2})$, when the typical digit sum begins with about $rj/2$ nines, for $j = 20, 40, 80$. The curves rise from the typical value to the peak in the same way for every j.

For the typical value, let D be uniform in $[10^j, 10^{j+1})$ and $\omega = \lceil \log_{10} j \rceil$. Writing $D = 2u$ or $2u+1$ gives $\lfloor 4.5D \rfloor = 9u$ or $9u+4$, and u runs over more than $10^j/2$ consecutive integers. Since $\gcd(9, 10) = 1$, the residue of $\lfloor 4.5D \rfloor$ modulo $10^{j-\omega}$ is within total variation $O(10^{-\omega})$ of uniform. So its digits in positions $h, \ldots, j-\omega$ are close to independent and uniform, and, as in step (G3) of the proof of Theorem 10.10, the sum of $\varphi(\{4.5D/10^i\})$ over these positions is $(j-h)/6 + O_P(\sqrt{j})$. The top ω positions contribute $O(\omega) = o(j)$. So $\beta_D \to \frac{1}{6}$ in probability and $\Phi(D) \to K_0(\frac{1}{6} + \frac{1}{6})^{-1/2} = K_0\sqrt{3}$. □

Remark 10.12 (Why Table 4 shows 1.4). The approach to the limit is logarithmically slow. At $D \approx 2 \cdot 10^7$ the random part of the digit sum has only about four digits and the deterministic part about four. The $O(1)$ boundary terms in $V(t)$ are then comparable with the main terms. Evaluating the main term by sampling 3000 digit sums from the Gaussian window, with exact integer arithmetic, at the peak $D = \lfloor 10^j/4.5 \rfloor$ and at the trough $D = \lfloor \frac{2}{9} y_j \rfloor$ gives ratios 1.517, 1.525, 1.531, 1.532, 1.545, 1.557, 1.558 and 1.566 for $j = 8, 12, 16, 24, 32, 48, 64, 96$. They increase towards $\sqrt{301}/11 = 1.5772$. The quantity $\rho(t)\sqrt{2\pi V(t)}$ of Theorem 10.9 averages 0.9865, 0.9948, 0.9982 and 0.9994 over random t with 10, 30, 100 and 300 digits.

11 Other bases

For an integer base $b \geq 2$ let $s_b(n)$ be the base-b digit sum and

$$f_b(n) = \begin{cases} n + s_b(n), & n \text{ odd}, \\ n - s_b(n), & n \text{ even}. \end{cases}$$

Write $\mathrm{C}_b(x, y)$ and $\mathrm{B}_b(x, y)$ for the numbers of carries and borrows in base b. The proof of Lemma 2.1 gives $s_b(x + y) = s_b(x) + s_b(y) - (b - 1)\,\mathrm{C}_b(x, y)$ and $s_b(x - y) = s_b(x) - s_b(y) + (b - 1)\,\mathrm{B}_b(x, y)$, and $s_b(n) \equiv n \pmod{b-1}$. The behaviour of f_b depends sharply on the parity of b.

Theorem 11.1 (Odd bases)**.** *If b is odd, every orbit of f_b reaches the fixed point 0; in particular f_b has no cycle other than $\{0\}$.*

Proof. For odd b, $b \equiv 1 \pmod 2$ gives $n \equiv s_b(n) \pmod 2$. If n is odd, then $s_b(n)$ is odd and $f_b(n) = n + s_b(n)$ is even. If $n > 0$ is even, then $s_b(n)$ is even and positive, so $f_b(n) = n - s_b(n)$ is even, non-negative and strictly smaller than n. Hence after at most one step the orbit consists of even numbers that decrease strictly until they reach 0. □

In an odd base, branching on the parity of n is the same as branching on the parity of $s_b(n)$, so f_b coincides with the base-b analogue of the iteration in OEIS A243260 [19]. The two rules differ only in even bases, and that is where all the structure lies.

Theorem 11.2 (Even bases: structure)**.** *Let $b \geq 2$ be even. Then, with 9 replaced by $b - 1$ and 10 by b throughout:*

1. *the parity of n is the parity of its last digit; for $b \geq 4$ every term after the first even term of an orbit is a multiple of $b - 1$;*
2. *$s_b((b-1)m) = (b-1)\,\mathrm{B}_b(bm, m)$, and on multiples of $b - 1$ the map is conjugate to $m \mapsto m \pm \kappa_b(m)$ with $\kappa_b(m) = s_b((b-1)m)/(b-1)$;*
3. *the level equation of Theorem 4.2 holds with $\kappa_b(k)$;*
4. *if $s_b(x)$ is even and $s_b(f_b(x) + b - 1) = s_b(x)$, then $f_b(f_b(x) + b - 1) = x + b - 1$;*
5. *an odd a lies on a 2-cycle if and only if $s_b(a)$ is odd and $s_b(s_b(a)) = (b-1)\,\mathrm{C}_b(a, s_b(a))$;*
6. *for a uniform on $\{0, \ldots, b^K - 1\}$, $\mathbb{E}\,\mathrm{C}_{b,K}(a, t) = \big(s_b(t) - t\,b^{-K}\big)/(b-1)$, and for a uniform on the odd numbers there and odd $t < b^K$, $\mathbb{E}\,\mathrm{C}_{b,K}(a, t) = \big(s_b(t) + 1 - (t+1)b^{-K}\big)/(b-1)$;*
7. *prefix lifting (Theorem 7.1) and its error bound (Corollary 7.2) hold with 10^L replaced by b^L and $9D$ by $(b-1)D$.*

Proof. The proofs for $b = 10$ use only the carry and borrow identities, the congruence $s_b(n) \equiv n \pmod{b-1}$, and two consequences of the evenness of b: the parity of n is that of its last digit, and $b - 1$ is odd, so that adding $b - 1$ swaps parity (used in (iv)). They therefore apply verbatim. In (vi), the odd digits $1, 3, \ldots, b - 1$ average $b/2$ and the even digits $0, 2, \ldots, b - 2$ average $b/2 - 1$, so $\mathbb{E}s_b(a) - \mathbb{E}s_b(a + t \bmod b^K) = 1$, and exactly $(t + 1)/2$ odd a overflow. All seven statements were also checked by computer for $b \in \{2, 4, 6, 8, 10, 16\}$. □

Proposition 11.3 (Resonance in base b)**.** *For even b and $j \geq 1$, the probability that adding $t = b^j - 1$ to a random odd integer produces exactly $s_b(t)/(b-1) = j$ carries is $(b-1)/b$.*

Proof. The units column gives $a_0+(b-1) \geq b$ because a_0 is odd, and each of the next $j-1$ columns gives $a_i + (b-1) + 1 \geq b$; so all j columns carry, and the count is exactly j iff column j does not carry, i.e. $a_j \neq b-1$. □

For $b \geq 4$ the switch lemma (Lemma 9.1) applies with the switch digits $t_i \in \{1, \ldots, b-2\}$. In base 2 there are no such digits, and the switch mechanism takes a different form.

Lemma 11.4 (Binary switch lemma)**.** *Let $t \geq 1$ have ℓ binary digits, let $0 < \delta \leq \frac{1}{2}$, and let a have independent bits with $\Pr(a_i = 1) \in [\delta, 1-\delta]$ for $1 \leq i \leq \ell$. Let*

$$S_2(t) = \{\, i \equiv 2 \pmod 4,\ 2 \leq i \leq \ell - 2 :\ t_{i-1} \neq t_i \,\}.$$

Then for every integer m, the number C of carries in $a+t$ (possibly truncated after any fixed column $k > \ell$) satisfies $\Pr(C = m) \leq K_1 \delta^{-2} |S_2(t)|^{-1/2} + e^{-\delta^2 |S_2(t)|/8}$.

Proof. For $i \in S_2(t)$ call i a *switch* if $a_{i-1} = t_{i-1}$ and $a_{i+1} = t_{i+1}$. The column triples $\{i-1, i, i+1\}$ for distinct $i \in S_2(t)$ are disjoint, so these events are independent, each of probability at least δ^2. Let W be the set of switches and condition on G, the bits outside $\{a_i : i \in W\}$; W is determined by G.

Let $i \in W$. In column $i-1$, $a_{i-1} + t_{i-1} = 2t_{i-1} \in \{0, 2\}$, so the carry into column i is $c_i = t_{i-1} = 1 - t_i$ whatever the carry into column $i-1$ (a *reset*). In column i, $a_i + t_i + c_i = a_i + 1$, so the carry out of column i is $\xi_i = a_i$. In column $i+1$, $a_{i+1} + t_{i+1} \in \{0, 2\}$, so the carry out of column $i+1$ is t_{i+1} whatever ξ_i is (a *block*). Hence a_i influences only the carry ξ_i, and as in Lemma 9.1, $C = C_0 + \sum_{i \in W} \xi_i$ with C_0 a function of G and the ξ_i independent Bernoulli variables with parameters in $[\delta, 1-\delta]$. The Kolmogorov–Rogozin inequality and the Chernoff bound for $|W|$ finish the proof exactly as in Lemma 9.1. □

Thus binary switches sit at the places where the bits of t change, and resonance in base 2 occurs when $t = 2^j - 1$ has no bit changes at all.

Theorem 11.5 (Two-cycle density in every even base)**.** *For every even $b \geq 2$ there are constants $c_b, K_b > 0$ and D_b such that for $D \geq D_b$*

$$\frac{c_b}{\sqrt{\log D}} \ \leq\ \frac{T_b(D)}{b^D} \ \leq\ \frac{K_b}{\sqrt{\log D}},$$

where $T_b(D)$ is the number of odd $a < b^D$ on a 2-cycle of f_b.

Proof. We go through the proofs of Theorems 9.4 and 10.5 and record each change. Fix an even b. Let a be uniform on $\{0, \ldots, b^D - 1\}$, with digit counts $N = (N_0, \ldots, N_{b-1})$ and $t = s_b(a)$, and let E be the event that a is the odd point of a 2-cycle. Let ℓ be the number of base-b digits of $(b-1)D$, so $t < b^\ell$, and set

$$h = \Big\lfloor \log_b \frac{\sqrt{D}}{\log D} \Big\rfloor, \qquad r = \Big\lceil \frac{\log D}{\log(4/3)} \Big\rceil, \qquad k = \ell + r.$$

Then $h \geq \frac{1}{4} \log_b D$ and $k \leq C_b \log D \leq C_b' h$ for large D. All constants below depend only on b.

(a) Characterisation. By Theorem 11.2(v),

$$E = \{a \text{ odd},\ t \text{ odd},\ \mathrm{C}_b(a, t) = m(t)\}, \qquad m(t) = \frac{s_b(t)}{b-1},$$

which is empty unless $(b-1) \mid s_b(t)$, i.e. $(b-1) \mid t$, since $s_b(t) \equiv t \pmod{b-1}$. In particular $E \subseteq \{\mathrm{C}_b(a, t) = m(t)\}$. Because b is even, the parity of a and of t is that of their last digit.

(b) Typical digit counts. Let A' be the event $\max_d |N_d/D - 1/b| \le \log D/\sqrt{D}$. By Hoeffding's inequality $\Pr(A'^c) = O_b(D^{-2})$. On A', every $p_d = N_d/D$ lies in $[\frac{1}{2b}, \frac{3}{4}]$ for large D, so we take $\delta = 1/(2b)$.

(c) Overflow beyond column $k-1$. The digits of t in columns $\ell, \dots, k-1$ are 0, so a carry passes column $k-1$ only if $a_\ell = \cdots = a_{k-1} = b-1$. Given $N \in A'$ this has probability at most $(p_{b-1} + k/D)^r \le (3/4)^r \le D^{-1}$.

(d) Low digits of t. The law of t is the D-fold convolution of the uniform law on $\{0, \dots, b-1\}$. It is log-concave, and by (2) its largest atom is $O_b(D^{-1/2})$. The proof of Lemma 9.3 then shows that $t \bmod M$ is within total variation $K_b M/\sqrt{D}$ of uniform. We use $M = b^h$ and $M = (b-1)b^h$; since $\gcd(b-1, b^h) = 1$, under the uniform law on $\mathbb{Z}/(b-1)b^h$ the residue mod $b-1$ and the last h digits are independent and uniform.

(e) Switch sets. For $b \ge 4$ let $S(t)$ be the set of odd $i < h$ with $t_i \in \{1, \dots, b-2\}$. The proof of Lemma 9.1 goes through word for word with 9 replaced by $b-1$. A switch is $i \in S(t)$ with $a_{i+1} + t_{i+1} \ne b-1$, which has probability at least $(b-1)\delta$, and the threshold $\theta_i = b - t_i - c_i$ lies in $\{1, \dots, b-1\}$. Under the uniform law on the last h digits, each odd position carries a digit in $\{1, \dots, b-2\}$ with probability $(b-2)/b \ge \frac{1}{2}$, so $|S(t)| \ge h/5$ except with probability $e^{-c_b h}$. For $b = 2$ we use $S_2(t)$ and Lemma 11.4. The bits t_{i-1}, t_i for distinct $i \equiv 2 \pmod 4$ are disjoint pairs of uniform bits, each pair unequal with probability $\frac{1}{2}$, so $|S_2(t)| \ge h/16$ except with probability e^{-ch}. In both cases the conditional decomposition

$$\mathrm{C}_{b,k}(a,t) = C_0 + \sum_{i \in W} \xi_i, \tag{6}$$

holds, with C_0 a function of the digits G outside W and, given G, independent Bernoulli ξ_i with parameters in $[\delta, 1-\delta]$. For $b = 2$, $\xi_i = a_i$. Moreover $|W| \ge c'_b h$ except with probability $e^{-c''_b h}$, by the Chernoff bound.

(f) Upper bound. Conditionally on $N \in A'$ with $|S(t)|$ (or $|S_2(t)|$) at least $h/16$, Lemma 9.2 (with b digit values; its proof does not use the number of values) replaces the first k digits by independent digits with law p at cost $k^2/(2D)$. The Kolmogorov–Rogozin inequality applied to (6) then gives $\Pr(\mathrm{C}_{b,k} = m) \le K_b/\sqrt{h}$. Together with (b)–(d),

$$\frac{T_b(D)}{b^D} \le O(D^{-2}) + e^{-c_b h} + \frac{K_b b^h}{\sqrt{D}} + \frac{K_b}{\sqrt{h}} + \frac{k^2}{2D} + D^{-1} = O_b\big((\log D)^{-1/2}\big).$$

(g) Lower bound. Let U' be the event that t is odd, $t \equiv 0 \pmod{b-1}$ and the switch set below h has at least $h/16$ elements. All three conditions depend only on $t \bmod (b-1)b^h$. By (d) and (e), $\Pr(U') \ge \frac{1}{2(b-1)}(1 - e^{-c_b h}) - K_b(b-1)b^h/\sqrt{D} \ge \frac{1}{4(b-1)}$ for large D. Fix $N \in A'$ with $t(N) \in U'$ and put $m = s_b(t)/(b-1)$, which is an integer. As in Step 2 of Theorem 10.5,

$$\Pr(E \mid N) \ge \Pr_p(a_0 \text{ odd})\, \Pr_p\big(\mathrm{C}_{b,k} = m \mid a_0 \text{ odd}\big) - \frac{k^2}{2D} - D^{-1},$$

and $\Pr_p(a_0 \text{ odd}) \ge \frac{1}{4}$.

Mean. Let $X = \mathrm{C}_{b,k}$ under $\Pr_p(\cdot \mid a_0 \text{ odd})$. By Theorem 11.2(vi), under uniform digits $\mathbb{E}X = m + \big(1 - (t+1)b^{-k}\big)/(b-1) \in [m, m+1]$. The change from uniform digits to law p moves the mean by at most $bk^2 \log D/\sqrt{D} = o(1)$, so $|\mathbb{E}X - m| \le 2$.

Variance. In the proof of Lemma 10.3, two runs of the carry recursion through column l disagree only if $a_l + t_l = b-1$, which has probability at most $1 - (b-1)\delta$. The lemma therefore holds in base b with $\operatorname{Var} X \le K_3 k$ and $K_3 = 1 + 2/((b-1)\delta)$. Column 0 again has an arbitrary law.

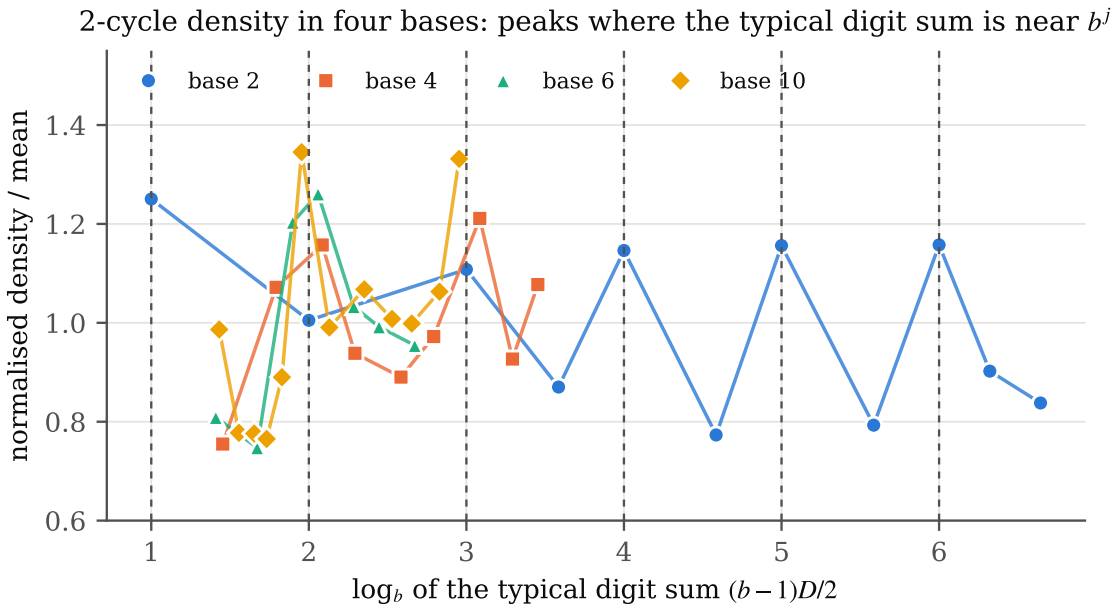


Figure 10: Exact normalised 2-cycle densities $\sqrt{\ln D}\, T_b(D)/b^D$ (each divided by its mean) in bases $2, 4, 6, 10$, plotted against $\log_b$ of the typical digit sum $(b-1)D/2$. In every base the peaks lie near integers, where the typical digit sum is close to $b^j - 1$ (Proposition 11.3).

Concentration and switches. Put $R = \max(2, 4\sqrt{K_3})$. Since $|\mathbb{E}X - m| \le 2 \le R\sqrt{k}/2$ for $k \ge 4$, Chebyshev's inequality gives $\Pr(|X - m| > R\sqrt{k}) \le \frac{1}{4}$. Steps 4 and 5 of the proof of Theorem 10.5 use only the decomposition (6), the bound $|W| \ge c'_b h$, and $k \le C'_b h$. Hence they apply unchanged, and Lemma 10.4 gives $\Pr_p(\mathrm{C}_{b,k} = m \mid a_0 \text{ odd}) \ge c_b/\sqrt{k}$.

Conclusion. For large D,

$$\frac{T_b(D)}{b^D} \ge \Pr(A' \cap U')\Big(\frac{c_b}{4\sqrt{k}} - \frac{k^2}{2D} - D^{-1}\Big) \ge \frac{c'_b}{\sqrt{\log D}}.$$

□

Figure 10 shows exact counts in four bases. In base 2 the normalised density oscillates between about 0.09 and 0.14, with peaks at $D = 16, 32, 64, 128$, where the typical digit sum $D/2$ is just below a power of 2, and troughs at $D = 24, 48, 96$. Theorems 11.1 and 11.5 together give a complete picture at the level of 2-cycles: in odd bases the dynamics is trivial, and in every even base the 2-cycles have density of exact order $(\log D)^{-1/2}$, with resonances located at the base-b repunits $b^j - 1$.

12 Convergence

We stress at the outset what is and is not proved. Every orbit computed so far is eventually periodic, but we do *not* prove that every orbit is eventually periodic (Conjecture 12.10). What we prove is an exact reformulation: eventual periodicity is equivalent to boundedness of the digit sums along the orbit (Theorem 12.2).

Lemma 12.1 (Crossing lemma)**.** *If a term $n < 10^J$ of an orbit is followed by $f(n) \geq 10^J$, then n is odd and $10^J - n \leq s(n)$; if moreover $s(n) \leq B < 10^t$, the leading $J - t$ digits of n are nines, so $s(n) \geq 9(J - t)$.*

Proof. If n were even, $f(n) = n - s(n) \leq n < 10^J$, a contradiction; so n is odd and $f(n) = n + s(n) \geq 10^J$, i.e. $10^J - n \leq s(n)$. If $s(n) \leq B < 10^t$ (with $t < J$), then $10^J - 10^t < 10^J - B \leq n < 10^J$. Every integer in the interval $(10^J - 10^t,\, 10^J)$ has exactly J digits, of which the leading $J - t$ are nines, since $10^J - 10^t$ is the number formed by $J - t$ nines followed by t zeros. Hence $s(n) \geq 9(J - t)$. □

Theorem 12.2 (Convergence criterion)**.** *The orbit of n is eventually periodic if and only if the digit sums of its terms are bounded. If they are bounded by $B < 10^t$, every term is below 10^J with $J = \max(\lceil \log_{10}(n + 1) \rceil, \lfloor B/9 \rfloor + t + 1)$.*

Proof. Eventually periodic orbits are finite. Conversely, $n < 10^J$; if a term reached 10^J, its predecessor m would satisfy $s(m) \geq 9(J - t) \geq 9(\lfloor B/9 \rfloor + 1) > B$ by Lemma 12.1. So the orbit is bounded, hence eventually periodic. □

Corollary 12.3 (Anatomy of a divergent orbit)**.** *A divergent orbit contains, for infinitely many J, an odd term $10^J - x$ with $1 \leq x \leq s(10^J - x)$: a block of nines followed by a short tail. Its digit sums are unbounded.*

Proposition 12.4 (Ascents with small digit sum)**.** *Let $9 \mid n_0$, n_0 odd, $s(n_0)$ even and $s(n_0) < 198$, and let $n_0 < n_1 < \ldots$ be the ascending run from n_0, i.e. the consecutive odd terms $n_{j+1} = n_j + s(n_j)$. Then the run is finite. Along it the digit sum never increases, except possibly at its last term, where it can exceed the previous digit sum by 9.*

Proof. Every odd n_j with $s(n_j)$ even is followed by the odd term $n_{j+1} = n_j + s(n_j)$. If $s(n_j)$ is odd, then $n_j + s(n_j)$ is even, so n_j is the last term of the run. Suppose $s(n_j) = S$ is even, a multiple of 9, and $18 \leq S \leq 180$, so $s(S) = 9$. By Lemma 2.1, $s(n_{j+1}) = S + 9 - 9C$ with $C = \mathrm{C}(n_j, S) \geq 0$. If C is odd, $s(n_{j+1}) \leq S$ is even and the run continues at a level that has not increased. If C is even, $s(n_{j+1})$ is odd, so n_{j+1} is the last term. Its digit sum is $S + 9$ if $C = 0$ and at most $S - 9$ otherwise.[1] By induction, every term before the last has an even digit sum of at most $s(n_0) \leq 180$, and the last term has digit sum at most $s(n_0) + 9 < 198$. The run is increasing with bounded digit sums, so it is finite by Lemma 12.1, as in the proof of Theorem 12.2. □

The next theorem removes the restriction to small levels for runs along which the level does not change. It applies the carry-balance idea along an arithmetic progression: while the level is constant, an ascent adds the same number $9k$ at every step, and the digits above the lowest ones then behave like a counter that must eventually pass a block of nines.

Theorem 12.5 (Constant-level runs are short)**.** *Let $9 \mid n_0$ have level $k \geq 1$, let ℓ be the number of digits of $9k$ and $\kappa = \kappa(k)$. Suppose $n_0, n_1, \ldots, n_r$ are consecutive terms of an orbit that all have level k. Then*

$$|n_r - n_0| < 2 \cdot 10^{\ell + \kappa} \leq 2 \cdot (90k)^2.$$

Proof. If k is odd, each step adds or subtracts the odd number $9k$, so parity alternates. An odd term n_j goes up to $n_j + 9k$, which is even, has level k and therefore returns to n_j; an even term likewise goes down and returns. So the run alternates between two values, and $|n_r - n_0| \leq 9k < 2 \cdot 10^{\ell}$.

If k is even, $9k$ is even and every term has the parity of n_0, so the run is either ascending ($n_{j+1} = n_j + 9k$, all n_j odd) or descending ($n_{j+1} = n_j - 9k$, all even); consider an ascending

[1] For example $8901 \to 8919$: $s = 18 \to 27$, after which $f(8919) = 8946$ is even and $8919 \leftrightarrow 8946$ is a 2-cycle.

run, the descending case being identical with borrows and zeros in place of carries and nines. By Theorem 4.2, every step produces exactly κ carries. Write $n_j = q_j 10^\ell + x_j$ with $0 \le x_j < 10^\ell$. Since $9k < 10^\ell$, each step increases q by 0 or 1, and by 1 exactly when it carries out of column $\ell - 1$; as x grows by $9k$ on every step that does not increase q, q increases at least once in every $\lceil 10^\ell/(9k) \rceil$ consecutive steps.

Suppose q increases at least 10^κ times during the run. The values $q_0, q_0 + 1, \ldots, q_0 + 10^\kappa - 1$ are then all taken, one of them, q^*, ends in κ nines, and the step that raises q from q^* to $q^* + 1$ belongs to the run. That step carries out of column $\ell - 1$ and then through the κ nines in columns $\ell, \ldots, \ell + \kappa - 1$, so it produces at least $\kappa + 1$ carries, a contradiction. Hence q increases at most $10^\kappa - 1$ times, so $r \le 10^\kappa \lceil 10^\ell/(9k) \rceil$ and

$$n_r - n_0 = 9k\, r \le 9k \cdot 10^\kappa \Big(\frac{10^\ell}{9k} + 1 \Big) = 10^{\ell+\kappa} + 9k \cdot 10^\kappa < 2 \cdot 10^{\ell+\kappa}.$$

Finally $\kappa \le \ell$ (the digit sum of $9k$ is at most 9ℓ) and $10^\ell \le 90k$, so $10^{\ell+\kappa} \le (90k)^2$. □

The bound was checked on 228 maximal constant-level runs in orbits of random numbers with 30 to 480 digits: none came within 14% of it. Since the level of a D-digit number is at most D, every constant-level run moves the orbit by $O(D^2)$, a negligible amount compared with n.

What is missing for global convergence. Theorem 12.5 does not by itself bound an *ascent*, i.e. a maximal stretch on the odd rail, because the level may change inside an ascent and stay even. For $k \le 10$ the only possible increase is to $k + 1$, which leaves the rail, and this gives Proposition 12.4. For larger k, $\kappa(k) \ge 2$ allows an increase by 2 that keeps the rail, and such increases do occur: among 448 ascents in orbits of random numbers with 50 to 800 digits we observed 14 level increases by 2 (and 244 by 1, at the end of an ascent). The heights of these ascents nevertheless grew only linearly with the number of digits (at most 39,240 for 800 digits). A proof of Conjecture 12.10 along these lines would need to show that level increases inside an ascent cannot accumulate, for example by controlling the level along the counter q above.

A computer-assisted proof for all starting values below explicit bounds

Prefix lifting turns Conjecture 12.10, restricted to a finite range, into a finite check, provided the blocks are large enough that a single constant-level run cannot cross one (Theorem 12.5). We state it for every even base b; recall that f_b has the parity of the last digit.

Theorem 12.6 (Block certificate)**.** *Let b be even, $D > L \ge 1$, $M = b^L$, $\sigma_{\max} = (b-1)(D-L)$ and $E = \sigma_{\max} + (b-1)L + 1$, and assume $\sigma_{\max} + (b-1)L < M$. For $0 \le \sigma \le \sigma_{\max}$ let $g_\sigma(x) = x \pm (\sigma + s_b(x))$ (plus for odd x, minus for even x) on $[0, M)$. Suppose that for every such σ no $x < E$ is carried by iterating g_σ to a value $\ge M$. Then every orbit of f_b starting at some $n < b^D$ stays below $(\lfloor n/M \rfloor + 2)M$; in particular it is bounded, hence eventually periodic, and* $\max(\mathit{orbit}) - n < 2M$.

Proof. Write each term as $H \cdot M + x$ with $0 \le x < M$. Since $s_b(HM + x) = s_b(H) + s_b(x)$ and $HM + x \equiv x \pmod 2$, as long as H is unchanged the low part evolves by $g_{s_b(H)}$; each step changes x by less than M (because $s_b(HM+x) \le \sigma_{\max} + (b-1)L < M$ whenever $s_b(H) \le \sigma_{\max}$), so H changes by at most 1 per step, upwards exactly when $g_{s_b(H)}$ leaves $[0, M)$ at the top. Right after such an upward step the new low part is $x' = y + s_b(HM + y) - M < s_b(HM + y) \le \sigma_{\max} + (b-1)L < E$.

Let $H_0 = \lfloor n/M \rfloor < b^{D-L}$. Suppose some term had high part $H_0 + 2$, and consider the first such term. Its predecessor has high part $H_0 + 1$, and the maximal stretch of the orbit at high part $H_0 + 1$ that contains it began with an upward step from H_0 (it cannot have begun with a downward

step from $H_0 + 2$, which has not yet occurred, nor with n itself). So that stretch starts at a low part $x' < E$ and leaves $[0, M)$ at the top under g_σ with $\sigma = s_b(H_0 + 1)$. As $H_0 + 1 \le b^{D-L}$ we have $\sigma \le \sigma_{\max}$, contradicting the hypothesis. Hence every term has high part at most $H_0 + 1$, so the orbit is contained in $[0, (H_0 + 2)M)$. A bounded orbit of a map on the integers is eventually periodic, and $\max(\text{orbit}) - n < (H_0 + 2)M - H_0 M = 2M$. □

The hypothesis is a finite computation: for each σ, the fate of every $x \in [0, M)$ under g_σ (absorbed at the top, absorbed at the bottom, or periodic) is determined exactly by iterating the functional graph $2^r \ge M + 2$ times with pointer doubling, in exact integer arithmetic.

Theorem 12.7 (Eventual periodicity below explicit bounds)**.** *Every orbit of the binary map f_2 starting below 2^{3024}, and every orbit of the decimal map f starting below 10^{341}, is eventually periodic. For these starting values,* $\max(\textit{orbit}) - n < 2 \cdot 2^{24}$ *and* $\max(\textit{orbit}) - n < 2 \cdot 10^8$*, respectively.*

Proof. The hypothesis of Theorem 12.6 was verified by computer for the parameters in Table 6. The last row of each base uses the climb method described below. □

base b	all $n <$	block b^L	$\sigma_{\max}$	entries checked E	time
2	2^{64}	2^{16}	48	65	1 s
2	2^{128}	2^{18}	110	129	3 s
2	2^{256}	2^{20}	236	257	28 s
2	2^{512}	2^{22}	490	513	4 min
10	10^{12}	10^5	63	109	1 s
10	10^{20}	10^6	126	181	14 s
10	10^{30}	10^7	207	271	3.5 min
2	2^{3024}	2^{24}	3000	3025	30 s*
10	10^{341}	10^8	2997	3070	20 s*

Table 6: Parameters for which the block certificate (Theorem 12.6) was verified, with running times on a laptop. Rows marked * use the climb method: every orbit is followed on the half-line and its maximum is compared with b^L.

Climb method. The hypothesis of Theorem 12.6 only concerns the finite low maps, and g_σ on $[0, b^L)$ agrees with the half-line map $x \mapsto x \pm (\sigma + s_b(x))$ until the orbit leaves the block. It therefore suffices to follow each entry $x_0 < E$ on the half-line, until it falls below 0 or closes a cycle, and to check that its maximum is below b^L. Memoising the maximum over each orbit makes this cost roughly linear in the number of states actually visited, rather than b^L. This removes the memory bottleneck of the block arrays. For $\sigma \le 3000$ and every entry below E, the highest point reached is 2,011,888 in base 10 and 2,107,392 in base 2, which gives the last two rows of Table 6. On the six small parameter sets used below, the climb method and the block method give identical verdicts, including the failure at $(2, 16, 8)$.

These ranges exceed exhaustive enumeration ($n \le 2 \cdot 10^6$) by a factor of about 10^{335} in base 10. Two checks guard against errors in the implementation. First, for small parameters ($(b, D, L) = (2, 18, 9), (4, 8, 5), (10, 5, 3), (10, 6, 4)$) every orbit starting below b^D was followed directly, and its high part never exceeded $H_0 + 1$, as the theorem asserts. Second, the certificate is not vacuous: for $(b, D, L) = (2, 16, 8)$ it fails, and direct computation shows that some orbit below 2^{16} does climb two blocks.

The block size needed grows only logarithmically: $L = 2\log_2 D + 4$ in base 2 for $D = 64, 128, 256, 512$, and $L = 5, 6, 7$ for $D = 12, 20, 30$ in base 10. The following lemma explains why a check at one scale controls larger scales. Write $\mathrm{Bad}_L(\sigma; W)$ if some $x < W$ reaches the top exit of $[0, b^L)$ under g_σ.

Lemma 12.8 (Scale reduction)**.** *Let $L > L' \geq 1$, b even, and $\sigma + (b-1)L \leq W \leq b^{L'}$. If $\mathrm{Bad}_L(\sigma; W)$, then $\mathrm{Bad}_{L'}(\sigma + j; W)$ for every $j \in \{0, 1, \ldots, (b-1)(L-L')\}$.*

Proof. Write $x = h\, b^{L'} + y$ with $0 \leq y < b^{L'}$. Then $s_b(x) = s_b(h) + s_b(y)$ and $x \equiv y \pmod 2$, so while h is fixed the orbit of g_σ follows $g_{\sigma + s_b(h)}$ on $[0, b^{L'})$. Every step is at most $\sigma + (b-1)L < b^{L'}$, so h changes by ± 1 at a time. An orbit starting at $x < W \leq b^{L'}$ (so $h = 0$) and leaving $[0, b^L)$ at the top must cross from h to $h+1$ for every $h < b^{L-L'}$. Fix such an h and consider the last entry into sub-block h before the first crossing to $h+1$. For $h = 0$ it is the start $y = x < W$. For $h \geq 1$ it comes from $h-1$, since $h+1$ has not been visited, and the entry point satisfies $y < \text{step} \leq \sigma + (b-1)L \leq W$. From y the map $g_{\sigma + s_b(h)}$ carries the orbit to the top of the sub-block, so $\mathrm{Bad}_{L'}(\sigma + s_b(h); W)$. As h ranges over $[0, b^{L-L'})$, $s_b(h)$ takes every value in $\{0, \ldots, (b-1)(L-L')\}$. □

So the bad set at scale L is contained in the set of σ whose whole window $[\sigma, \sigma + (b-1)(L-L')]$ is bad at scale L'. A single good parameter at a small scale therefore certifies $(b-1)(L-L') + 1$ parameters at every larger scale. Numerically, though, the bad set at a fixed scale is not sparse. Once σ exceeds roughly $b^{L/2}$, almost every σ is bad, so the lemma speeds up the computation but does not by itself give a proof.

The quantity that governs the problem is $L_{\min}(\sigma)$, the least L with no $x < \sigma + (b-1)L + 1$ exiting at the top. Every $\sigma \leq \theta_L$ satisfies $L_{\min}(\sigma) \leq L$, where

b	10	10	10	10	2	2	2	2	2
L	3	4	5	6	10	12	14	16	18
θ_L	69	179	963	1952	54	113	244	495	1010
b^L/θ_L^2	0.21	0.31	0.11	0.26	0.35	0.32	0.28	0.27	0.26

In both bases the block needed is quadratic in σ, with $b^{L_{\min}} \approx \sigma^2/4$. This is the order of the height $2b^{\ell+\kappa} \leq 2((b-1)bk)^2$ of a constant-level run in Theorem 12.5. It suggests that climbs from the bottom are no higher than single constant-level runs, so the level changes inside an ascent do not compound. Conjecture 12.10 would therefore follow from:

For the statement we use the half-line map $\bar{g}_\sigma(x) = x + \sigma + s_b(x)$ (x odd), $x - \sigma - s_b(x)$ (x even), on $\mathbb{Z}_{\geq 0}$, with orbits stopped when they become negative. It does not involve a block length; g_σ on $[0, b^L)$ agrees with it until the orbit leaves the block.

Conjecture 12.9 (Quadratic climbs)**.** *There is an absolute constant c such that for every even b, every $\sigma \geq 0$ and every $X \geq 0$, every orbit of $\bar{g}_\sigma$ that starts at some $x_0 \leq X$ stays below $c\,(\sigma + X + 1)^2$.*

Conjecture 12.9 implies Conjecture 12.10. Given D, choose L minimal with $b^L > c\,(2(b-1)D + 1)^2$, so $L = O(\log D)$, and apply Theorem 12.6. Its hypothesis concerns $\sigma \leq \sigma_{\max} = (b-1)(D-L)$ and entries $x_0 < E = \sigma_{\max} + (b-1)L + 1$. With $X = E$ the conjecture bounds every such orbit by $c(\sigma_{\max} + E + 1)^2 \leq c(2(b-1)D + 1)^2 < b^L$, so no orbit reaches the top of the block. The data of Theorem 12.7 are consistent with $c = 1$: for $\sigma \leq 3000$ and $X = E$ the highest point is 2,011,888 $< (3000 + 3073)^2$ in base 10, and 2,107,392 $< (3000 + 3025)^2$ in base 2. For $10 \leq \sigma \leq 3000$ and all entries $x \leq \sigma + (b-1) \cdot 8$ (base 10) or $x \leq \sigma + 24$ (base 2), about 4.7 million starting points in each base, the highest point is below $1.71(\sigma + 1)^2$ in base 10 and $1.19(\sigma + 1)^2$ in base 2. The number of separate up-runs in a climb is not bounded, however. It reaches 19 in base 10 and 61

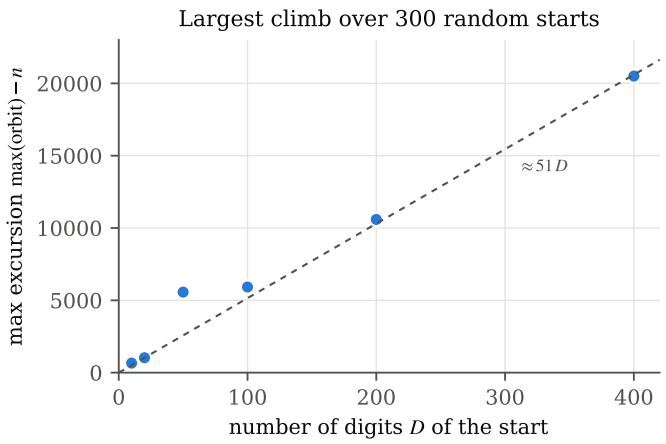


Figure 11: Largest excursion over 300 random starting values for each number of digits.

in base 2, with up to 33 and 61 level changes. So a proof cannot simply count runs: it must show that successive runs do not stack. It is a statement about the finite maps g_σ alone. Theorem 12.5 proves it for runs at constant level. What remains is to control level increases inside an ascent, which we do not know how to do.

The structure above explains convergence heuristically: regular steps are translations that Theorem 5.1 pairs into ascents and mirrored descents, so net drift can only come from exceptional carries, which require long blocks of nines. Numerically, every orbit from $n \le 2 \cdot 10^6$ is eventually periodic, the largest excursion $\max(\text{orbit}) - n$ there is 777 (at $n = 998457$), and for 300 random starts (Python `random.Random(0)`) with each of $10, \dots, 400$ digits the excursion grows roughly linearly in the number of digits (Figure 11).

Conjecture 12.10. *Every orbit of f is eventually periodic; equivalently, digit sums are bounded along every orbit.*

Remark 12.11. Theorem 7.1 suggests a computer-assisted proof for $n < 10^D$: track the low map g_σ and show that the prefix cannot climb indefinitely. A coarse version (after the prefix increases it cannot increase again before decreasing) fails for sparse clusters of σ; a finer version tracking the last digits of H and the side from which each level is entered remains to be carried out.

13 Cycle statistics

Two different quantities are easy to confuse here: the number of *distinct cycles* of a given length, and the proportion of *starting values* whose orbit enters a cycle of that length. They can differ substantially, because basins have very different sizes. Table 8 and Figure 13 give both.

Table 7 counts the distinct cycles reached from multiples of 9 up to N (each cycle counted once, regardless of the point at which it is entered) and the number of steps before the orbit enters its cycle.

Table 8 and Figure 12 list the cycles found. The statement in the introduction that 2-cycles dominate refers to the number of distinct cycles; by basin size they account for about seven-tenths of all starting values. Odd lengths occur; which lengths are possible, and how the turns produce them, is not yet classified. In the 15-cycle the ascent adds 18 six times and 27 once and the descent subtracts 18 seven times and 9 once, both totalling 135.

N	distinct cycles	mean steps	std. steps	max steps
500	20	0.47	0.78	3
1,000	26	1.95	2.97	13
10,000	171	1.35	1.70	13
100,000	2,037	1.16	1.44	13
200,000	4,149	1.16	1.51	27

Table 7: Orbits of the multiples of 9 up to N: number of distinct cycles reached, and statistics of the number of steps before entering the cycle (exact enumeration).

length	#cycles	% of cycles	% of starts	smallest example
2	37374	86.5	72.3	9, 18
3	81	0.19	0.70	8991, 9018, 9000
4	4759	11.0	20.7	2871, 2889, 2916, 2898
5	72	0.17	0.57	289953, 289989, 290034, 290016, 289998
8	605	1.40	2.64	89793, 89829, …, 89820
14	17	0.04	0.20	89991, 90027, …, 90000
15	284	0.66	2.79	1791, 1809, …, 1899, 1926, …, 1800
16	18	0.04	0.16	189783, 189819, …, 189810

Table 8: Cycles other than the fixed point $\{0\}$ reached from $1 \le n \le 2 \cdot 10^6$: number of distinct cycles, their share of all 43,210 cycles, the share of starting values n whose orbit enters a cycle of that length (only the six starts $1, 2, 3, 4, 6, 8$ reach $\{0\}$), and the cycle of smallest minimum for each length.

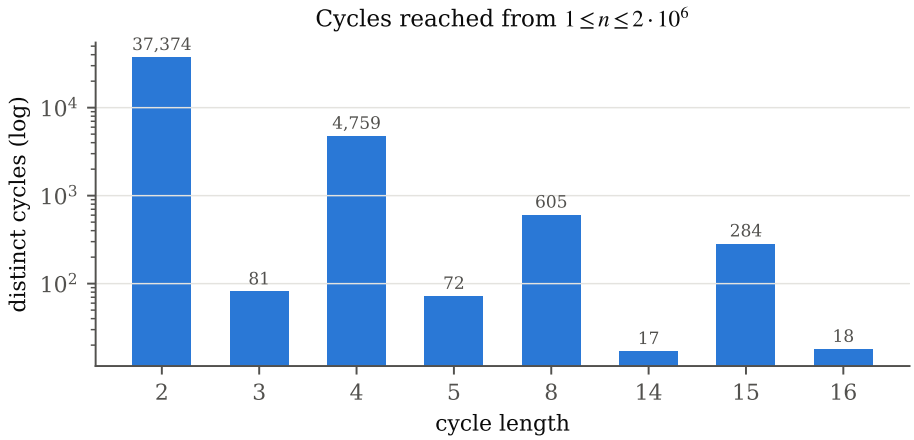


Figure 12: Distribution of cycle lengths (logarithmic scale).

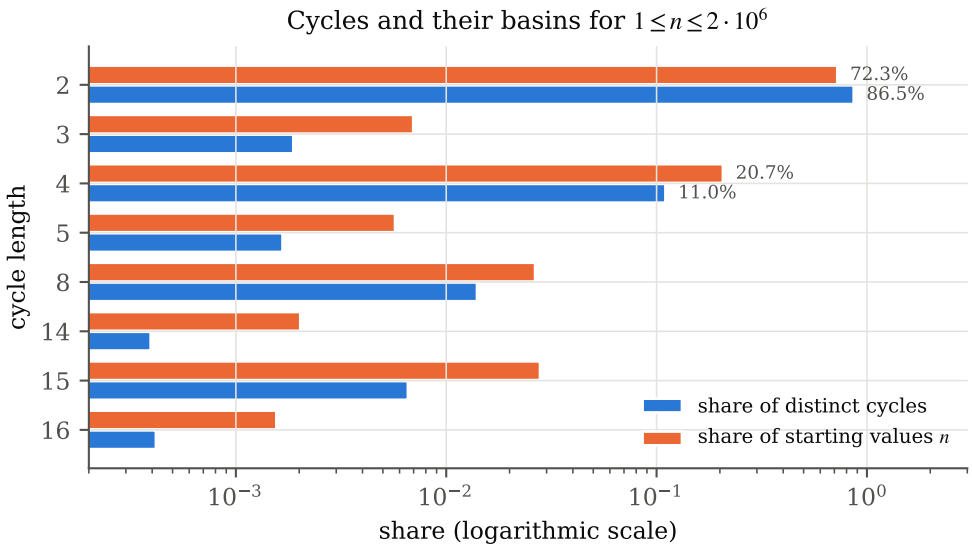


Figure 13: Share of distinct cycles (blue) and of starting values (orange) by cycle length, for $1 \le n \le 2 \cdot 10^6$. The 2-cycles are 86.5% of the cycles but attract only 72.3% of the starts; 4-, 8- and 15-cycles have larger basins than their numbers suggest.

14 Open problems

1. Prove Conjecture 12.10. It is proved below 10^{341} (Theorem 12.7); by Theorem 12.6 the general case would follow from the absence of upward crossings for the finite maps g_σ with blocks of size $b^{O(\log \sigma)}$. By Theorem 12.5 it would also suffice to show that level increases inside an ascent cannot accumulate. The same question can be asked in every even base; base 2, where the digit sum is the number of ones, may be the most tractable case.

2. Corollary 10.11 determines the limit points of the normalised 2-cycle density. Extend the oscillation law to every even base; the analogue of Lemma 10.6 holds with 10 replaced by b, and the extremal problem of Lemma 10.7 changes with b.

3. Do *all* cycles, not only 2-cycles, have density zero? In the census, 2-cycles are about 86% of all *distinct* cycles, but points on longer cycles are not rare. Exhaustively, 5.0%, 5.2% and 5.2% of $n \le N$ are periodic for $N = 10^5, 10^6, 10^7$, and the share of them on 2-cycles falls from 73% to 60%. Sampling 40,000 random D-digit numbers gives periodic proportions of 5.7%, 4.5%, 4.9%, 4.4%, 5.0%, 4.3% and 4.2% (standard error 0.1%) for $D = 8, 16, \dots, 512$, with 4-cycles about as frequent as 2-cycles and apparently oscillating with D. The data therefore neither confirm nor refute density zero. The natural conjecture is that the periodic proportion is of order $(\log D)^{-1/2}$, like that of 2-cycles, which would make the decay invisible at these sizes. For a 4-cycle $a \to b \to c \to d \to a$ with two ascending and two descending steps, closure means $s(a) + s(b) = s(c) + s(d)$; by Lemma 2.1 this is a linear condition on three carry and borrow counts, $\mathrm{C}(a, s(a))$, $\mathrm{C}(b, s(b))$ and $\mathrm{B}(c, s(c))$, which depend on overlapping low digits. The switch argument of Lemma 9.1 handles one such count; a proof for 4-cycles needs a *joint* anti-concentration bound for these coupled counts.

4. Determine the profile $c_L(\sigma)$ analytically, and classify the possible cycle lengths.

5. Explain the rare cycles with two ascending runs.

Data and code

All computations are reproducible with the accompanying Python package (`digitdyn`), archived at `[repository URL and DOI to be added]`. It contains: the exhaustive orbit and cycle census (Proposition 5.2, Tables 8 and 7); the digit dynamic programme for $T(D)$ (Table 3), tested against exhaustive enumeration for $D \le 5$; the prefix-lifting counter (Theorem 7.1, Corollary 7.2); the carry-resonance computation (Theorem 9.7); exact checks of Theorems 3.1, 4.2, 5.1 and 10.1; a test suite; and a single script that regenerates every table and figure. Throughout the paper we distinguish *theorems* (proved), *computational statements* (exact computations over a stated finite range) and *conjectures*.

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